

Active Inference for Intelligent Traffic Light Control

An Active Inference Framework for Adaptive
Traffic Light Control

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Abstract

Introduction Traffic systems are increasingly dynamic and uncertain, making fixed-time and rule-based controllers less suitable for modern mobility demands. This thesis examines whether active inference, a probabilistic and uncertainty-aware control framework, can support intelligent traffic light control.

Research Question The study investigates whether active inference can match or exceed reinforcement learning and rule-based methods in efficiency, adaptability, and robustness.

Method Using design science research and simulation-based evaluation, a multi-agent active inference controller was implemented in SUMO. It was compared with a rule-based controller and a Deep Q-Network (DQN) under non-stationary and uncertain traffic conditions, using vehicle idle time, CO₂ emissions, bus priority service rate, and phase-switching frequency.

Results The active inference controller achieved the lowest vehicle idle time and CO₂ emissions across all scenarios. Under sensor noise, both DQN and active inference outperformed the rule-based baseline on efficiency metrics, but DQN degraded under noise, weather, and traffic disturbances. The results also show a trade-off: DQN achieved the highest bus priority service rate, while active inference had the highest phase-switching frequency.

Discussion The findings suggest that adaptability to uncertainty is critical for effective traffic signal control. While reinforcement learning performed well under controlled conditions, active inference showed stronger resilience in non-stationary and partially observable environments, highlighting its potential as a robust alternative for intelligent traffic control.

Keywords: Active Inference; Intelligent Traffic Light Control; Adaptive Traffic Control; SUMO; Deep Q-Network; Reinforcement Learning; Non-Stationary Traffic; Expected Free Energy

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Abbreviations

Abbreviation	Description
AIF	Active Inference
AI	Artificial Intelligence
CO ₂	Carbon Dioxide
CPU	Central Processing Unit
CTM	Cell Transmission Model
DQN	Deep Q-Network
EFE	Expected Free Energy
FEP	Free Energy Principle
GPU	Graphics Processing Unit
KL	Kullback–Leibler Divergence
MPC	Model Predictive Control
RL	Reinforcement Learning
SCATS	Sydney Coordinated Adaptive Traffic System
SCOOT	Split Cycle Offset Optimization Technique
SUMO	Simulation of Urban MObility

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1

Introduction

1.1 Context and Motivation

Urban traffic congestion is more than a daily inconvenience, it has significant consequences in a wide range of areas [37]. Research indicates that traffic-related air pollution negatively affects human health [26, 35], with well documented impacts linked specifically to emissions from congested road networks. Heavy traffic also slows economic activity [42] and contributes to road accidents [45], further underscoring the societal importance of effective traffic management strategies.

To address these challenges, modern cities rely on a range of traffic control mechanisms [31], among which traffic light control plays a central role in regulating flow and mitigating congestion. However, traditional control methods often struggle to adapt to real-time fluctuations in traffic demand [23], unexpected events, and complex urban dynamics. As urban mobility patterns become increasingly dynamic, the limitations of static or rule-based control strategies become more pronounced. This motivates the development of more intelligent and adaptive traffic light control systems. In an increasingly automated world, characterized by AI-driven systems and autonomous decision-making, there is a growing need for controllers capable of responding to real-time traffic conditions, congestion patterns, road closures, accidents, and other dynamic variables.

Most contemporary approaches to traffic light control rely on deep reinforcement learning, which has become the dominant paradigm in recent years [34, 50]. Earlier methods include rule-based strategies, representing one of the simplest classes of control strategies [8, 9]. More advanced alternatives are optimization-based techniques [51], which incorporate predictive modeling to improve decision making [52]. These approaches typically rely on explicit traffic flow models and solve optimization problems at each control step, allowing them to anticipate future states rather than reacting solely to current observations.

Despite decades of research and widespread deployment, traffic light control in real-world environments remains dominated by relatively simple strategies. Most intersections still operate using fixed-time schedules derived from historical averages or rule-based logic triggered by basic sensor inputs [49]. These systems are robust and easy to maintain.

1.2 Problem

Real traffic systems are highly non-stationary: demand fluctuates across time, unexpected events occur frequently, and sensor data is often noisy or incomplete. The problem of this study is that existing traffic-light control methods struggle to operate reliably under such variability and uncertainty. Traditional rule-based controllers rely on predefined, human designed rules to determine signal behavior, for example, simple “if X happens, do Y” logic. While easy to implement, these methods lack the flexibility required to handle the variability and unpredictability of modern urban traffic and are not well suited for complex or rapidly changing real-world scenarios [17]. Deep reinforcement learning has shown strong performance in controlled settings, but RL-based traffic signal control remains difficult to transfer to real systems because of training and tuning demands, sensing limitations, non-stationary traffic conditions, and the limited transparency of neural policies [4, 27]. As a result, RL-based controllers remain rare outside controlled pilot deployments.

This gap between research and real-world deployment highlights the need for control frameworks that can operate under uncertainty, adapt to

changing conditions, and remain interpretable and robust. Active inference originates from computational neuroscience, where it is used to describe how agents perceive, infer, and act under uncertainty. AIF offers such a perspective by explicitly modeling uncertainty, learning generative models of the environment, and selecting actions that minimize expected free energy [32]. Yet its potential for traffic light control remains largely unexplored, creating an opportunity to investigate whether AIF can provide a viable alternative in settings dominated by uncertainty, partial observability, and non-stationarity.

1.3 Research Questions

This thesis investigates whether AIF can serve as an effective framework for intelligent traffic-light control and how it compares to established approaches. The study focuses on evaluating performance, adaptability, and robustness within a simulated environment that reflects real-world complexity. The research is guided by the following questions:

RQ1: How does AIF compare to representative state-of-the-art traffic-light control methods, such as deep reinforcement learning and rule-based controllers, in terms of traffic efficiency, stability, and overall flow optimization?

RQ2: How well does an AIF controller adapt to non-stationary or fluctuating traffic conditions, including sudden demand shifts, incidents, and stochastic disturbances?

RQ3: What are the trade-offs of using AIF compared to existing traffic-light control methods in terms of interpretability, computational feasibility, and suitability for real-world deployment?

Together, these questions evaluate whether AIF can match or exceed existing approaches in improving traffic efficiency (RQ1), whether it can adapt effectively to dynamic and uncertain environments where traditional controllers and many RL methods struggle (RQ2), and whether it offers practical advantages or disadvantages regarding transparency, computational demands, and integration into existing traffic-management systems (RQ3).

1.4 Scope and Delimitations

This thesis is delimited to a simulation-based evaluation of AIF for adaptive traffic light control. It does not aim to validate AIF in real traffic infrastructure, but examines the behaviour of an AIF controller under controlled and progressively uncertain simulation conditions.

The experimental setting is limited to a SUMO-based four-arm intersection. This scope allows a consistent comparison between controllers, but excludes larger traffic networks, multiple interacting intersections, and integration with existing traffic-management infrastructure.

The comparison is limited to a rule-based baseline, a DQN agent, and an AIF controller. Other approaches, such as model predictive control, Proximal Policy Optimization, Advantage Actor-Critic, and larger multi-agent reinforcement learning systems, are outside the scope of this thesis.

The evaluation is limited to four metrics: vehicle idle time, CO₂ emissions, bus priority service rate, and phase switching frequency. Pedestrian flow, safety validation, economic cost, driver behaviour, and operator acceptance are not evaluated.

2

Extended Background

This chapter provides the theoretical and methodological foundations necessary to understand the problem of intelligent traffic light control and the framework explored in this thesis. It surveys established approaches, outlines the characteristics of real-world traffic dynamics, and introduces the principles of AIF that motivate the proposed control method.

2.1 Intelligent Traffic Light Control

Intelligent traffic light control refers to a class of adaptive light control strategies that use real time data, computational models, and automated decision making to optimize traffic flow under dynamic and uncertain conditions. Unlike fixed-time or rule-based systems, which rely on predefined timing plans or simple sensor triggered logic, intelligent controllers continuously adjust signal phases in response to evolving traffic states. Their primary objective is to improve mobility, reduce congestion, and enhance overall network efficiency while accounting for different traffic dynamics [47].

A defining characteristic of intelligent traffic light control is its reliance on real-time perception of the traffic environment. This typically involves sensor networks, camera systems, or simulated observations that provide estimates of queue lengths, vehicle arrivals, delays, and flow patterns [7].

Since traffic measurements are often incomplete or subject to noise, these controllers must make decisions based on imperfect information [2], leveraging adaptive strategies and learning-based approaches to optimize signal timing. Another key characteristic is adaptivity. Intelligent controllers are designed to respond to fluctuations in demand, such as peak hour surges, incident induced congestion, or irregular traffic patterns caused by weather or roadworks. Adaptivity distinguishes intelligent systems from static timing plans, which assume stable and predictable conditions. Reinforcement learning and model predictive control exemplify this adaptive capability by updating actions based on current observations or predicted future states [34, 52].

The overarching goal of intelligent traffic light control is to optimize traffic performance across multiple dimensions. Common objectives include minimizing average delay, reducing queue lengths, increasing throughput, lowering stop frequency, and improving travel time reliability. In some cases, broader societal goals, such as reducing emissions, improving safety, or prioritizing public transport are incorporated into the control objective. Because these goals may conflict, intelligent controllers often balance multiple performance criteria simultaneously [43]. Evaluating whether these goals are achieved requires well-defined performance metrics. Standard metrics include average vehicle delay, queue length, intersection throughput, travel time, and the number of stops per vehicle [22]. More advanced evaluations may incorporate fuel consumption or emission estimates, especially in sustainability-oriented studies [43]. In multi-intersection networks, coordination metrics such as progression efficiency or bandwidth utilization are also relevant. These metrics allow for systematic comparison between intelligent controllers and baseline methods such as fixed time, actuated, or rule-based systems [22, 31].

Finally, intelligent traffic light control systems must operate within the constraints of real-world traffic infrastructure. This includes hardware limitations, communication delays, sensor reliability, and safety requirements. As a result, practical implementations often combine predictive modeling, online learning, and robust decision making to ensure stable and interpretable behavior. Recent research emphasizes the importance of controllers that can adapt to non-stationary traffic conditions, handle uncertainty, and integrate seamlessly with existing traffic management systems [47]. In summary, intelligent traffic light control represents a shift from

static, rule-based decision making toward adaptive, data driven, and uncertainty aware control. Its defining characteristics, real-time perception, adaptability, predictive reasoning, and multi-objective optimization—make it a natural application area for AIF, which explicitly models uncertainty, preferences, and future outcomes within a unified probabilistic framework [14].

2.2 Existing Approaches to Traffic Light Control

2.2.1 Fixed-Time and Actuated Control

Traditional traffic light control relies on predefined timing plans designed using historical traffic data. Fixed-time controllers, such as Webster’s method [48] and coordinated signal plans in TRANSYT [36], operate according to static phase schedules. Actuated controllers incorporate real-time sensor inputs, commonly loop detectors, to trigger phase changes based on rule-based logic, as implemented in SCOOT [19] and SCATS [28]. These methods are typically designed offline [31]; actuation uses online measurements to execute rule-based decisions rather than learn or optimize in real-time. They assume relatively stable traffic patterns and reliable sensor measurements.

2.2.2 Optimization-Based Approaches

Optimization-based approaches formulate light control as a mathematical problem in which variables such as phase durations, cycle lengths, or offsets are chosen to minimize objectives like delay, queue length, or stop frequency. Representative methods include linear and nonlinear programming, mixed-integer optimization, and queue-based models [31]. These methods are typically solved offline using predicted traffic states or historical data, though some approaches allow limited online re-optimization. They rely on the assumption that traffic dynamics can be reasonably approximated over the optimization horizon.

2.2.3 Model Predictive Control

Model Predictive Control (MPC) applies a receding horizon strategy, repeatedly solving an optimization problem based on predicted traffic evolution under candidate control actions. MPC controllers often use store-and-forward models or cell transmission models to predict queue dynamics while minimizing congestion-related costs under operational constraints. MPC operates online, updating control actions at each time step using current observations and predictions, but it relies on a model that is designed or trained offline [1].

2.2.4 Reinforcement Learning Approaches

Reinforcement learning frames traffic light control as a sequential decision-making problem in which an agent learns a control policy by maximizing expected cumulative reward. Representative algorithms include Q-learning, DQN, policy gradient methods, and actor-critic methods [49]. Multi-agent RL approaches are often used in multi-intersection settings to coordinate decentralized controllers [5]. Training is typically performed offline in simulation due to sample complexity, but learned policies can be deployed online, sometimes with limited online adaptation. RL assumes a Markovian state representation and a reward function that reflects traffic performance metrics such as delay or queue length.

2.3 Characteristics of Traffic Dynamics

Traffic systems present substantial modeling challenges due to their inherent complexity and variability. Traffic dynamics are fundamentally non-deterministic, exhibiting temporal patterns that vary significantly across different periods, such as peak hour versus off-peak conditions [20]. Furthermore, traffic flow is subject to numerous exogenous factors that lie outside standard modeling assumptions, including adverse weather conditions, traffic incidents, roadwork activities, and other unforeseen disruptions [30].

Accurate characterization of traffic states depends critically on sensor networks, which provide imperfect and noisy observations. Measurement distortions can arise from occlusions, environmental conditions, sensor

malfunctions, or data processing errors, rendering the true traffic state only partially observable. Additionally, driver behavior introduces stochastic elements into the system, as individual driving decisions exhibit considerable variability and unpredictability [20].

From a statistical perspective, these factors contribute to non-stationary system dynamics, where the distributional properties of both observations and state transitions evolve over time. This non-stationarity poses significant challenges for modeling and control, as assumptions about fixed or time-invariant system dynamics may not hold consistently under shifting conditions [20]. Given these characteristics, traffic light control can benefit from modeling frameworks that explicitly account for uncertainty and allow for adaptation to evolving system dynamics, rather than assuming static, pre-specified operational regimes.

2.4 Active Inference: Theoretical Foundation

AIF is a theoretical framework for understanding how biological agents perceive and act in the world, grounded in the free energy principle from theoretical neuroscience. At its core, it proposes that agents maintain internal probabilistic models of the world and continuously update these models to remain in states they expect to occupy. In this context, surprise, which is formally defined as the negative log probability of an observation given an agent's model of the world $-\log p(o_t | s_t)$, reflects how unexpected an outcome is. AIF is founded on the principle that an agent acts to minimize surprise about its observations over time. Because surprise itself cannot be evaluated directly, the agent instead minimizes an upper bound on surprise known as variational free energy [10]. In this framework, perception corresponds to updating beliefs to better explain observations, while action is selected to make future observations more predictable or desirable.

The agent maintains a generative model of its environment, consisting of latent (hidden) states s , observations o , and actions a . The generative model comprises a likelihood mapping $P(o|s)$, which specifies the probability of observations given states, and transition dynamics $P(s'|s, a)$, which describes how states evolve over time conditioned on actions. Prior beliefs over initial states are encoded as $P(s)$, while the agent encodes preferences over observations in the form of a prior distribution $P(o)$, where observations with higher prior probability correspond to preferred outcomes

[13].

Action selection is guided by the minimization of expected free energy, which evaluates future policies based on two complementary considerations. On one hand, actions are favored if they are expected to lead to observations that align with the agent’s preferences, a contribution often referred to as pragmatic value. On the other hand, actions may be selected because they are expected to reduce uncertainty about the hidden states of the environment, a contribution known as epistemic value [11].

Formally, the expected free energy (EFE) of a policy can be decomposed into two components: a *pragmatic* term and an *epistemic* term.

The pragmatic term captures how well expected observations align with the agent’s preferences. Intuitively, it encourages actions that are expected to yield preferred outcomes. Mathematically, this is expressed as the expected negative log probability of preferred observations under the predictive distribution:

$$\text{Pragmatic term} = \mathbb{E}_{q(o)}[-\ln p(o)],$$

where o denotes observations, $p(o)$ represents the prior preference distribution over observations, and $q(o)$ is the predictive distribution over future observations.

The epistemic term quantifies the expected information gain about hidden states of the environment. It encourages actions that reduce uncertainty about the hidden states s . This is measured by the Kullback–Leibler divergence between posterior beliefs given observations and prior beliefs:

$$\text{Epistemic term} = \mathbb{E}_{q(o)}[D_{\text{KL}}(q(s | o) \| q(s))],$$

where $q(s)$ is the prior belief over hidden states, and $q(s | o)$ is the posterior belief after observing o .

Together, these components define the expected free energy G of a policy:

$$G = \mathbb{E}_{q(o)}[-\ln p(o)] - \mathbb{E}_{q(o)}[D_{\text{KL}}(q(s | o) \| q(s))],$$

which the agent seeks to minimize when selecting actions.

2.5 Active Inference in Control

After defining AIF and the quantity it seeks to minimize, we can describe how policy selection operates in a control setting. Under AIF, policies are evaluated according to their expected free energy, and the optimal policy is formally defined as the one that minimizes this quantity [11]

$$\pi^* = \arg \min_{\pi} G(\pi).$$

This expression describes an idealized form of policy selection in which the agent would deterministically choose the policy with the lowest expected free energy. In practice, however, expected free energy is evaluated under uncertainty, as it depends on predicted future observations and approximate beliefs derived from an imperfect generative model. As a result, purely greedy policy selection can be overly sensitive to estimation noise and may lead to premature commitment to suboptimal behaviors [14]. For this reason, policy selection in AIF is typically stochastic. Rather than selecting only the single best policy, the agent maintains a probability distribution over policies, assigning higher probability to policies with lower expected free energy. This distribution is commonly defined using a softmax function [13, 32].

$$q(\pi) = \frac{\exp(-\beta G(\pi))}{\sum_{\pi'} \exp(-\beta G(\pi'))}.$$

The inverse temperature parameter β controls the sharpness of the policy distribution. Higher values yield more deterministic behavior, increasingly approximating greedy selection, while lower values promote exploration by flattening the distribution over policies [12]. Importantly, stochastic policy selection preserves the epistemic component of expected free energy minimization, allowing the agent to occasionally select informative policies that reduce uncertainty about the environment and support online learning and adaptation.

Although these findings support the use of AIF in control problems, existing work also shows important limitations. Most AIF applications have been evaluated in controlled domains such as robotics and artificial agents, where the environment, action space, and observation model can be defined more precisely than in urban traffic systems [24]. This makes it difficult to assume that results from other control domains transfer directly

to traffic light control, where demand patterns, disturbances, and sensing quality vary over time. Similarly, reinforcement learning studies report strong performance in simulation, but their dependence on reward design, training exposure, and stable state representations can limit transfer to uncertain or partially observable traffic environments. Therefore, the present study treats AIF as a promising but still underexplored control framework rather than as an already validated alternative for real-world traffic management.

3

Method

3.1 Method Choice

This study adopts Design Science Research (DSR) as its overall research strategy. DSR is suitable for studies that develop and evaluate artefacts intended to address identified problems in a specific domain [18, 21]. In this thesis, the artefact is an AIF-based traffic light controller. The study therefore combines design research, through the construction of the controller, with empirical research, through controlled simulation-based evaluation.

The choice of DSR is justified by the research question, since the thesis does not only analyse existing traffic-control methods but designs, implements, and evaluates a new controller. DSR provides a methodological structure for connecting the research problem to artefact design, demonstration, and evaluation [33, 21]. This structure is appropriate because the controller is assessed through measurable performance outcomes rather than theoretical argumentation alone.

The empirical evaluation is conducted as a controlled simulation experiment. This strategy is appropriate because real-world testing of an unvalidated traffic light controller would be difficult to control and could create safety risks. Simulation makes it possible to expose all controllers to the

same traffic demand profiles, disturbance settings, and evaluation scenarios, supporting a consistent comparison between the rule-based controller, the DQN agent, and the AIF controller [6].

The data collection method consists of automated logging of simulation outputs from SUMO through Traffic Control Interface (TraCI). The collected data include vehicle idle time, CO₂ emissions, bus priority service rate, and phase switching frequency. The data analysis method is quantitative and based on descriptive comparison of controller performance across scenarios. This distinction between research strategy, data collection method, and data analysis method follows standard methodological terminology [6].

3.1.1 Design Science Research Process

The study follows the process model proposed by Peffers et al. [33], consisting of six stages: problem identification and motivation, definition of solution objectives, design and development, demonstration, evaluation, and communication. This process was selected because it fits the design-oriented nature of the thesis. The study develops an artefact in the form of an AIF-based traffic light controller and evaluates its behaviour in a controlled simulation environment. The Peffers et al. process provides a clear structure for connecting the traffic-control problem to the design objectives, implementation, demonstration, and evaluation. It also supports methodological transparency by making each step of the artefact development and assessment explicit.

1. Problem identification and motivation

The problem of adaptive traffic light control under uncertainty is established through a structured literature review of existing light control strategies, including fixed-time, optimisation-based, and learning-based approaches. Literature synthesis is used as an analytical method to identify limitations in handling stochastic arrivals, partial observability, and unexpected disturbances. This stage corresponds to DSR's requirement that artefact development be grounded in a clearly articulated and theoretically informed problem domain [18].

2. Define the objectives of a solution

The objectives of the solution are derived from the principles of traffic flow and the AIF framework. They are expressed through measurable outcomes: vehicle idle time, CO₂ emissions, bus priority service rate, and phase switching frequency. These metrics reflect the study’s focus on traffic efficiency, environmental impact, public transport priority, and signal stability. Since the study is technical and simulation-based, the requirements are defined through conceptual modelling and analytical reasoning rather than stakeholder elicitation.

3. Design and development

The artefact is constructed according to the theoretical principles of AIF, including generative model formulation, belief updating mechanisms, and action-selection policies. Development follows an iterative refinement logic, consistent with DSR’s emphasis on progressive improvement of artefacts [18]. Intermediate versions are tested and adjusted to align the implementation with the defined objectives.

4. Demonstration

The artefact is demonstrated within a microscopic traffic simulation environment. This demonstration stage follows the DSR process, where the artefact is applied to show how it can address the identified problem [33]. A simulation-based setting is appropriate because real-world testing of an unvalidated traffic light controller would be difficult to control and could create safety risks [6]. In this thesis, SUMO provides the microscopic simulation environment used to represent vehicle-level dynamics, stochastic disturbances, and partially observable traffic conditions [3]. This setup makes it possible to demonstrate the controller under identical and repeatable experimental conditions before considering real-world deployment.

5. Evaluation

Evaluation is conducted through controlled simulation-based experimentation. Disturbance conditions are systematically varied, while vehicle idle time, CO₂ emissions, bus priority service rate, and phase switching frequency are measured. The AIF controller is compared with the rule-based controller and the DQN agent under identical experimental conditions. This structure supports a systematic comparison of controller behaviour across scenarios and satisfies DSR’s requirement for artefact assessment through measurable criteria [33].

6. Communication

The research process, design decisions, and evaluation outcomes are documented transparently to enable critical scrutiny, replication, and future extension.

3.1.2 Alternative Research Strategy

An alternative research strategy would have been a controlled simulation experiment without using DSR as the overarching framework. Experimental research is suitable when outcomes are compared under controlled conditions by varying selected factors and measuring their effects [6]. This strategy could answer the same research questions, since the rule-based controller, the DQN agent, and the AIF controller could still be evaluated under the same SUMO scenarios, traffic demand profiles, and disturbance conditions, using the same metrics: vehicle idle time, CO₂ emissions, bus priority service rate, and phase switching frequency.

This strategy was not selected as the main research strategy because the thesis also develops an artefact: an AIF-based traffic light controller. DSR is more suitable because it connects problem identification, design objectives, artefact development, demonstration, evaluation, and communication [33, 21]. The controlled simulation experiment is therefore used as the evaluation part of the study, while DSR better reflects the full research process.

A more analytical alternative would be a deductive modelling strategy based on operations research or control theory. Queueing-theoretic and max-pressure control formulations exemplify this tradition [1, 44, 25]. However, this strategy is less suitable for the present research questions, because they require comparing controller behaviour under stochastic disturbances, partial observability, and non-stationary traffic demand rather than deriving theoretical guarantees.

3.1.3 Ethical Considerations in Design Science Research

Ethical considerations in DSR concern both the research process and the potential societal implications of the artefact [18]. In this study, experimentation is conducted entirely within a virtual simulation environment.

This eliminates risks to human participants and avoids unintended consequences that could arise from deploying unvalidated AI-based control systems in real traffic settings.

From a research ethics perspective, the study follows principles of transparent reporting, reproducibility, and responsible innovation. Transparent reporting and reproducibility are important because the artefact, simulation settings, parameters, and evaluation procedure must be described clearly enough to support scrutiny and replication [6, 21]. Responsible innovation is relevant because AI-based traffic control could affect safety, mobility, and public infrastructure if deployed outside simulation [39].

Because the study is conducted entirely in simulation and does not involve human participants, personal data, or sensitive personal data, it falls outside the type of research requiring ethical review under the Swedish Ethical Review Act [40, 41]. Nevertheless, the research acknowledges broader ethical implications of AI-based infrastructure control, including issues of reliability, safety, and unintended bias in decision-making systems. By restricting evaluation to controlled simulation and clearly communicating methodological limitations, the study aligns with ethical standards relevant to artefact-oriented engineering research.

3.1.4 Methodological Transparency and Reproducibility

This study is designed to make its research process transparent and reproducible by clearly outlining the methodological logic behind the development and evaluation of the proposed controller. The emphasis is on how the study is structured, its research strategy, experimental logic, and evaluation flow, rather than on implementation details.

The subsequent sections of the Method chapter provide the concrete elements required for replication, including artefact specifications, simulation settings, parameter values, baseline configurations, and evaluation procedures. Together, these components allow independent researchers to reconstruct the experimental environment and reproduce the empirical results.

The validity, reliability, generalizability, and objectivity of the study are discussed in Chapter 5, where the implications of the simulation-based research design are evaluated.

3.2 Environment

The simulation environment is implemented in Simulation of Urban Mobility (SUMO), a microscopic traffic simulation platform [3]. It consists of a single four-arm intersection with incoming and outgoing lanes in each direction. Vehicles enter the network from all approaches and traverse the intersection according to predefined routes.

The environment includes multiple vehicle types, including standard passenger vehicles and buses. Buses are treated as priority vehicles and are used to evaluate the ability of controllers to prioritise public transport.

To model realistic sensing conditions, observation noise is introduced through occlusion effects, where larger vehicles may obscure smaller ones, leading to imperfect detection of traffic states. This noise can be further amplified under adverse weather conditions, which degrade sensor reliability.

Additionally, stochastic traffic incidents such as accidents can be introduced, temporarily blocking lanes and disrupting normal traffic flow. These mechanisms allow the environment to capture key sources of uncertainty present in real-world traffic systems.

The physical structure of the simulated environment is illustrated in Figure 3.1, which shows the layout of the four-arm intersection.

Types of Vehicles on the Road

To model heterogeneous traffic dynamics, we consider three vehicle types: passenger cars, buses, and trucks. These are evaluated along three dimensions: CO₂ emissions, public priority, and perceptual ambiguity. Passenger cars have low emissions, no priority, and do not affect sensor reliability. Buses are assigned public priority, produce moderate emissions, and reduce sensor reliability due to potential occlusion of surrounding vehicles. Trucks have no priority, generate high emissions, and similarly reduce sensor reliability.

This categorization allows us to introduce environmental cost, normative preference, and epistemic uncertainty into the AIF framework.

Traffic Dynamics

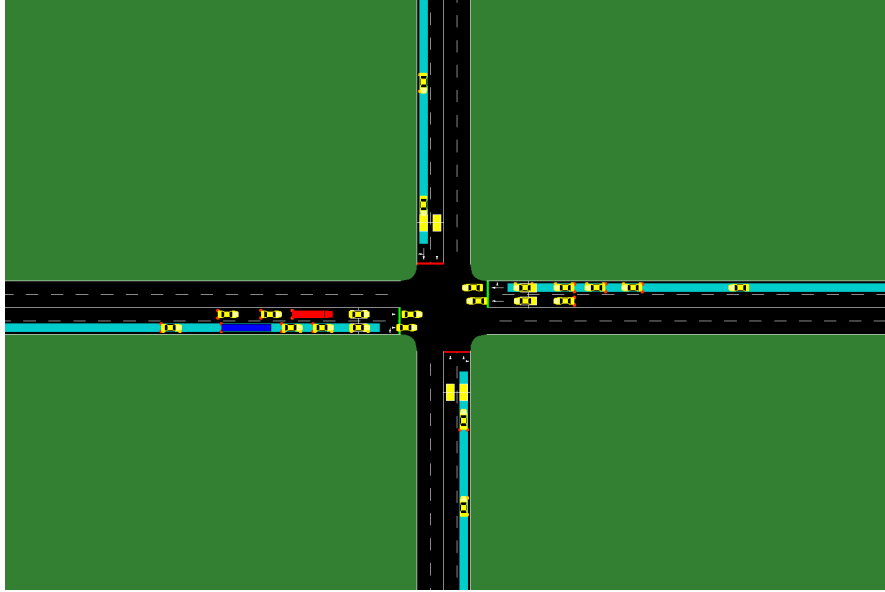


Figure 3.1: SUMO intersection layout.

Table 3.1: Vehicle-type properties used in the heterogeneous traffic model; CO₂ values are derived from simulation logs.

Vehicle Type	CO ₂ (mg/veh)	Public Priority	Sensor Ambiguity Impact
Passenger Car	≈ 2990	None	None
Bus	≈ 11,519	High	High (occlusion)
Truck	≈ 12,922	None	High (occlusion)

To approximate realistic urban traffic dynamics, we model a full-day demand profile with time-varying vehicle flow. Traffic volume begins at a low level during early hours, increases rapidly to a high morning peak, transitions to a moderate steady flow during midday, rises again during the evening rush hour, and subsequently decreases exponentially during late-night hours.

This non-stationary demand structure allows us to evaluate the agent’s ability to adapt to changing traffic regimes over time. This is particularly meaningful because the agent has no explicit information about time of day encoded in its internal model, meaning it cannot anticipate demand shifts in advance and must instead rely solely on real-time observations to update its beliefs and adapt its behavior accordingly.

3.2.1 Stochastic Environmental Events: Weather and Accidents

To evaluate the robustness of the proposed AIF-based traffic control system, the simulation environment incorporates stochastic external disturbances. These disturbances are designed to mimic real-world non-stationarities such as adverse weather conditions and traffic accidents, which significantly affect traffic dynamics and sensor reliability.

Weather-Induced Observation Degradation

A time-dependent weather process is introduced to simulate reduced visibility and degraded sensing performance. Weather conditions are modeled as a binary latent variable $w \in \{0, 1\}$, where $w = 1$ indicates adverse weather (e.g., heavy rain or fog), and $w = 0$ represents normal conditions.

At a predefined time interval $[t_{start}, t_{end}]$, adverse weather is activated, and a visual disturbance is introduced into the simulation environment. More importantly, weather directly affects the observation model through a multiplicative reliability factor γ^w , where $\gamma \in (0, 1)$ reduces the probability of correctly detecting vehicles under adverse conditions.

This mechanism ensures that during bad weather, the agent receives noisier and less reliable observations, thereby increasing epistemic uncertainty and testing the inference capabilities of the controller.

Traffic Accident Injection

To further increase environmental stochasticity, a traffic accident event is injected at a randomly selected time step. The accident is modeled as a sudden disruption affecting a single vehicle, which becomes immobile and acts as a temporary obstruction in the traffic flow.

The victim vehicle is selected from the set of active vehicles in the simulation at the time of the event. To ensure realistic congestion impact, the selection prioritizes vehicles located on the East-West (EW) lanes. Among these vehicles, the one closest to the intersection is selected based on lane position. Once selected, the vehicle's velocity is set to zero, effectively simulating a blockage: The vehicle remains stationary for a fixed duration before

being removed from the simulation, representing clearance of the accident scene.

These disturbances are essential for evaluating whether the controller can maintain stable performance under realistic, unpredictable conditions. In particular, they test the ability of the AIF framework to adapt its belief updates and policy selection under increased epistemic uncertainty.

3.2.2 Observation Noise and Occlusion Modeling

To ensure realism in the perception process of the traffic control system, the agent does not observe the true vehicle counts directly. Instead, observations are generated through a stochastic corruption model that accounts for sensor noise, environmental effects (e.g., weather), and partial occlusion of vehicles, particularly in congested traffic conditions.

Let N_d^{true} denote the true number of vehicles in direction $d \in \{NS, EW\}$. The observed vehicle count N_d^{obs} is generated through a two-stage stochastic process.

Weather-Dependent Observation Reliability

First, observation reliability is modulated by an environmental factor representing weather conditions. A binary variable $w \in \{0, 1\}$ indicates normal and adverse weather conditions respectively. The observation success probability is defined as

$$p_d = \gamma^w \cdot (1 - \alpha \cdot \max(1, N_d^{true} - 1)) , \quad (3.1)$$

where $\gamma \in (0, 1)$ controls the impact of weather on sensor reliability. Close 0 values heavily deteriorate readings and cause the agent to severely undercount the vehicles, while the closer it is to 1, the less severe the weather effect is. The parameter α is a scaling factor that increases the uncertainty of the observation as the density of the traffic increases. This formulation captures the empirically observed phenomenon that dense traffic reduces detection accuracy due to occlusion and sensor overlap.

Stochastic Observation Model

Given the reliability-adjusted probability p_d , the preliminary observed count is sampled using a binomial process:

$$\tilde{N}_d \sim \text{Binomial}(N_d^{true}, p_d) \quad (3.2)$$

This formulation models imperfect detection where each vehicle is independently observed with probability p_d .

Occlusion in Heavy Traffic

In high-density conditions, occlusion effects become more pronounced, particularly when the number of trucks and buses dominates the traffic composition. To model this, an additional occlusion term is introduced:

$$O_d = \sum_{i=1}^{K_d} o_i, \quad o_i \sim \mathcal{D}_{occ} \quad (3.3)$$

where K_d is the number of heavy vehicles (buses and trucks), and $\mathcal{D}_{occ}$ is a discrete distribution over $\{0, 1, 2, 3\}$ with probabilities:

$$P(o_i = x) = \begin{cases} 0.50 & \text{if } x = 0 \\ 0.30 & \text{if } x = 1 \\ 0.15 & \text{if } x = 2 \\ 0.05 & \text{if } x = 3 \end{cases} \quad (3.4)$$

Occlusion is activated only under congestion conditions, specifically when:

$$N_d^{true} \geq 3 \cdot K_d \quad (3.5)$$

This ensures that occlusion is applied only when passenger vehicles dominate and heavy vehicle density is sufficiently high to plausibly obstruct sensing.

Final Observation Model

The final observed vehicle count is computed as:

$$N_d^{obs} = \max(0, \tilde{N}_d - O_d) \quad (3.6)$$

This stochastic observation process makes the environment partially observable, thereby motivating the use of belief-based and inference-driven control strategies.

Road Usage

The road network consists of a main road and a smaller secondary road, both following the same daily traffic dynamics. The secondary road, however, experiences systematically lower vehicle flow and a reduced probability of trucks and buses compared to the main road. This design creates an imbalanced scenario, allowing us to evaluate how the agent adapts when one road is heavily prioritized over the other.

3.3 Control Strategies

Three controllers are evaluated in this study: a rule-based baseline, a DQN agent, and an AIF controller. The rule-based controller uses fixed heuristics based on observed traffic counts and bus presence. The DQN agent is trained in simulation before evaluation and then tested under the same scenario conditions as the other controllers. The AIF controller operates using a specified generative model, manually defined transition probabilities, and preference-based action selection. During evaluation, the active inference controller updates its beliefs over hidden traffic states but does not perform online learning of model parameters.

All three controllers were evaluated under reasonable default configurations. For the DQN agent, standard best practices were applied, including experience replay, a target network, and an ε -greedy exploration strategy. However, the reward function was intentionally kept simple and was not extensively engineered, reflecting a practical deployment perspective where the goal is to assess how each method performs under reasonable conditions rather than under heavily optimized ones. Extensive reward shaping is rarely feasible in real-world traffic control settings, and evaluating methods without it provides a more honest basis for comparison.

3.4 Active Inference Agent

3.4.1 Observation Model

We first analyzed the observed traffic data using the rule-based agent as a baseline to find meaningful empirical traffic patterns. Vehicle counts, CO₂ emissions, and bus presence were observed across the simulation and empirically binned into six regimes spanning from very low traffic to traffic jam conditions. For each bin, the corresponding mean vector and covariance matrix were computed directly from the observed data, yielding the state-specific parameters used in the generative model. These regimes were defined to ensure interpretability and computational tractability. It should be noted that the state space was derived from data collected under the rule-based controller, which is a reasonable choice as rule-based controllers represent the most commonly deployed approach in real-world traffic systems, making them a natural baseline for characterizing typical traffic conditions.

Accordingly, the hidden state variable s_i , $i = 1, \dots, 6$, represents these discretised traffic conditions.

For each hidden state, we assume that the corresponding observations consist of a three-dimensional vector $\mathbf{o} \in \mathbb{R}^3$ following a multivariate normal distribution:

$$\mathbf{o} \mid s_i \sim \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i), \quad i = 1, \dots, 6,$$

where $\boldsymbol{\mu}_i$ and $\boldsymbol{\Sigma}_i$ denote the state-specific mean vector and covariance matrix, respectively. This distributional assumption was made for pragmatic and methodological reasons. A multivariate normal likelihood provides a tractable way to model continuous observations while still capturing correlations between vehicle counts, CO₂ emissions, and bus presence through the covariance matrix. However, it is also a simplification. Real traffic observations may be skewed, bounded, discrete, or affected by rare incident-driven spikes that are not fully captured by a Gaussian likelihood. The assumption is therefore used as an interpretable approximation for belief updating, not as a claim that real traffic observations are exactly normally distributed.

Specifically, the parameters are given by:

$$\begin{aligned}\boldsymbol{\mu}_1 &= \begin{bmatrix} 2.98 \\ 7.85 \\ 0.52 \end{bmatrix}, & \boldsymbol{\mu}_2 &= \begin{bmatrix} 9.13 \\ 22.39 \\ 1.17 \end{bmatrix}, & \boldsymbol{\mu}_3 &= \begin{bmatrix} 15.16 \\ 33.17 \\ 1.94 \end{bmatrix}, \\ \boldsymbol{\mu}_4 &= \begin{bmatrix} 21.25 \\ 49.07 \\ 2.63 \end{bmatrix}, & \boldsymbol{\mu}_5 &= \begin{bmatrix} 28.28 \\ 60.12 \\ 2.78 \end{bmatrix}, & \boldsymbol{\mu}_6 &= \begin{bmatrix} 39.05 \\ 73.24 \\ 2.59 \end{bmatrix},\end{aligned}$$

$$\begin{aligned}\boldsymbol{\Sigma}_1 &= \begin{bmatrix} 3.62 & 10.03 & 0.58 \\ 10.03 & 45.91 & 1.30 \\ 0.58 & 1.30 & 0.47 \end{bmatrix}, & \boldsymbol{\Sigma}_2 &= \begin{bmatrix} 2.60 & 4.62 & 0.30 \\ 4.62 & 50.99 & -0.45 \\ 0.30 & -0.45 & 0.91 \end{bmatrix}, \\ \boldsymbol{\Sigma}_3 &= \begin{bmatrix} 3.15 & 6.43 & 0.25 \\ 6.43 & 80.84 & 0.41 \\ 0.25 & 0.41 & 1.30 \end{bmatrix}, & \boldsymbol{\Sigma}_4 &= \begin{bmatrix} 3.33 & 11.59 & -0.34 \\ 11.59 & 125.50 & -4.22 \\ -0.34 & -4.22 & 1.72 \end{bmatrix}, \\ \boldsymbol{\Sigma}_5 &= \begin{bmatrix} 1.98 & 4.81 & -0.52 \\ 4.81 & 79.47 & 1.34 \\ -0.52 & 1.34 & 1.71 \end{bmatrix}, & \boldsymbol{\Sigma}_6 &= \begin{bmatrix} 25.68 & 30.18 & 0.36 \\ 30.18 & 102.20 & 0.16 \\ 0.36 & 0.16 & 1.72 \end{bmatrix}.\end{aligned}$$

These multivariate normal distributions are then used to compute the likelihood of noisy observations within the Active Inference framework.

3.4.2 Transition Probabilities

The transition probabilities for the lanes were manually defined to capture plausible traffic dynamics under red and green phases. Manual specification was chosen here as a deliberate design decision to maintain interpretability and transparency of the generative model. This is consistent with the broader goal of AIF as an inspectable control framework, where the modeler can directly encode domain knowledge about how traffic states evolve. The matrices were validated through preliminary testing to ensure they produced stable and reasonable controller behavior. Let the hidden traffic states be denoted as

$$\text{States} = \{\text{null}, \text{low}, \text{medium}, \text{high}, \text{extra_high}, \text{jam}\}.$$

The transition matrices are defined as follows:

Red phase for any lane

$$\mathbf{T}_{\text{red}} = \begin{bmatrix} 0.40 & 0.55 & 0.04 & 0.01 & 0.00 & 0.00 \\ 0.05 & 0.30 & 0.55 & 0.09 & 0.01 & 0.00 \\ 0.00 & 0.00 & 0.20 & 0.70 & 0.08 & 0.02 \\ 0.00 & 0.00 & 0.00 & 0.20 & 0.60 & 0.20 \\ 0.00 & 0.00 & 0.00 & 0.00 & 0.70 & 0.30 \\ 0.00 & 0.00 & 0.00 & 0.00 & 0.10 & 0.90 \end{bmatrix},$$

where each entry $T_{\text{red}}[i, j]$ represents the probability of transitioning **from state i to state j** during a red phase.

Green phase for any lane:

$$\mathbf{T}_{\text{green}} = \begin{bmatrix} 0.80 & 0.18 & 0.02 & 0.00 & 0.00 & 0.00 \\ 0.15 & 0.60 & 0.22 & 0.03 & 0.00 & 0.00 \\ 0.08 & 0.32 & 0.44 & 0.14 & 0.02 & 0.00 \\ 0.02 & 0.08 & 0.33 & 0.42 & 0.13 & 0.02 \\ 0.01 & 0.03 & 0.33 & 0.43 & 0.15 & 0.05 \\ 0.05 & 0.10 & 0.25 & 0.30 & 0.20 & 0.10 \end{bmatrix},$$

where each entry $T_{\text{green}}[i, j]$ represents the probability of transitioning **from state i to state j** during a green phase.

These matrices were designed to reflect the intuitive behavior of traffic: during a red phase, vehicles tend to accumulate, increasing the likelihood of transitioning to higher congestion states, while during a green phase, vehicles tend to leave, increasing the probability of moving toward lower congestion states.

3.4.3 Preference, Pragmatic, and Epistemic Value Computation

Preference scores To quantify how desirable each hidden state is, we defined a set of preference scores based on a target (preferred) distribution over observations. Specifically, we assumed a preferred multivariate normal distribution with mean μ^* and covariance Σ^* , representing desirable traffic conditions. For this simulation, the preferred mean vector was set to

$$\boldsymbol{\mu}^* = \begin{bmatrix} 5 \\ 14 \\ 0 \end{bmatrix} \quad (3.7)$$

and the preferred covariance matrix was set to that of the low congestion state

$$\boldsymbol{\Sigma}^* = \boldsymbol{\Sigma}_1 = \begin{bmatrix} 3.62 & 10.03 & 0.58 \\ 10.03 & 45.91 & 1.30 \\ 0.58 & 1.30 & 0.47 \end{bmatrix} \quad (3.8)$$

This choice encodes a preference for low vehicle counts, low CO₂ emissions, and minimal bus congestion. The preferred mean vector was deliberately set between the first and second state mean vectors, reflecting a preference for reasonably low traffic conditions without targeting near-empty roads. This ensures the agent is not incentivised to push already low-traffic lanes further toward zero, which would be an unrealistic and counterproductive control objective. The preferred covariance matrix is set to that of the low congestion state, encoding the expectation that desirable conditions exhibit similarly low variance in observations.

For each hidden state s_j , characterized by its own observation model $\mathcal{N}(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$, we computed a preference score using the log-likelihood of $\boldsymbol{\mu}_j$ under the preferred distribution. This results in:

$$\log P(s_j) = -\frac{1}{2} \left(d \log(2\pi) + \log |\boldsymbol{\Sigma}^*| + \text{Tr}((\boldsymbol{\Sigma}^*)^{-1} \boldsymbol{\Sigma}_j) + (\boldsymbol{\mu}^* - \boldsymbol{\mu}_j)^\top (\boldsymbol{\Sigma}^*)^{-1} (\boldsymbol{\mu}^* - \boldsymbol{\mu}_j) \right),$$

where d is the dimensionality of the observation space. This formulation is closely related to the Kullback–Leibler divergence between Gaussian distributions and captures both differences in means and uncertainty.

The resulting scores were normalised to ensure numerical stability and comparability across states.

The pragmatic value for each candidate action was computed by weighting preference scores with the predicted next-state distribution, and further scaled by the relative observed traffic load in the North-South and East-West directions to account for asymmetric demand:

$$PV_{NS} = w_{NS} \cdot \text{score}_{NS} + w_{EW} \cdot \text{score}_{EW} \quad (3.9)$$

where weights are proportional to observed vehicle counts. The epistemic value was computed as the KL divergence between the predicted next-state distribution and the prior belief, encouraging actions that lead to informative state transitions. All values were normalised across actions before computing the EFE.

3.4.4 Decision-Making and Belief Updates

Expected free energy

The decision-making process is based on the Expected Free Energy (EFE), which combines both pragmatic (goal-directed) and epistemic (information-seeking) components. For each candidate action a , the EFE was computed as:

$$\text{EFE}(a) = -PV(a) - \lambda EV(a),$$

where $PV(a)$ denotes the pragmatic value, $EV(a)$ the epistemic value, and λ is a weighting coefficient controlling the importance of exploration. In our implementation, $\lambda = 0.5$. This value was chosen to weight pragmatic value more strongly than epistemic value, reflecting the primary objective of minimising congestion rather than maximising information gain. Epistemic exploration is still preserved but treated as secondary to goal-directed behavior.

Concretely, for the two possible traffic light actions (green for North-South and green for East-West), we compute:

$$\text{EFE}_{NS} = -PV_{NS} - 0.5 \cdot EV_{NS},$$

$$\text{EFE}_{EW} = -PV_{EW} - 0.5 \cdot EV_{EW}.$$

Minimizing EFE corresponds to selecting actions that both achieve desirable outcomes and reduce uncertainty about the system.

Action selection

Actions were selected probabilistically using a softmax function over the negative expected free energy values

$$p(a) = \frac{\exp(-\beta \cdot \text{EFE}(a))}{\sum_{a'} \exp(-\beta \cdot \text{EFE}(a'))},$$

where β is an inverse temperature parameter controlling the stochasticity of decisions. In this work, $\beta = 3$ was chosen through minor preliminary testing. It provides sufficient sharpness to favour lower EFE actions while still preserving stochastic exploration, avoiding the degenerate case of fully deterministic action selection.

The action (traffic light phase) was then sampled according to this distribution. This introduces controlled randomness, preventing the controller from becoming overly deterministic and allowing adaptation to changing traffic conditions.

Belief update

After selecting an action, the prior belief over hidden states was updated using the predicted next-state distribution corresponding to the chosen action:

$$q_{t+1}(s) = q_t(s' \mid a),$$

where $q_t(s' \mid a)$ is the predicted state distribution under the selected action. The updated belief was then normalized:

$$q_{t+1}(s) \leftarrow \frac{q_{t+1}(s)}{\sum_s q_{t+1}(s)}.$$

This updated distribution serves as the prior for the next time step, enabling sequential belief propagation over time.

Simple Rule-Based Traffic Controller

For comparison with the trained agents, a simple rule-based traffic light controller was implemented. The controller relies on fixed heuristics and does not perform any learning or prediction. It operates at discrete evaluation intervals of $\Delta t = 10$ simulation seconds. At each evaluation step, it uses noisy vehicle counts obtained from the simulation (which may

be affected by weather, occlusion, or accident events) to decide whether to maintain or switch the current green phase.

The controller aggregates traffic into two directions: north–south (NS) and east–west (EW). For each evaluation step, it computes a priority score for both directions:

$$S_{\text{NS}} = O_{\text{NS}} + \beta_{\text{bus}} \cdot B_{\text{NS}}, \quad (3.10)$$

$$S_{\text{EW}} = O_{\text{EW}} + \beta_{\text{bus}} \cdot B_{\text{EW}} + \beta_{\text{main}}, \quad (3.11)$$

where

- O_{NS} and O_{EW} are the observed (noisy) vehicle counts in the NS and EW directions, respectively;
- B_{NS} and B_{EW} denote the number of buses waiting in each direction;
- $\beta_{\text{bus}} = 5$ is a fixed bonus applied per waiting bus;
- $\beta_{\text{main}} = 2$ is a constant bias favoring the EW direction, which is treated as the main road.

To prevent rapid oscillations between phases, the controller enforces both a minimum and a maximum green duration, measured in evaluation steps. Let k denote the number of consecutive evaluations for which the current phase has been active. The implementation uses

$$k_{\min} = 2, \quad k_{\max} = 6.$$

The values were chosen to balance responsiveness and stability. A minimum duration of 2 evaluation steps prevents rapid phase switching caused by short-term traffic fluctuations, while a maximum duration of 6 steps ensures that no traffic direction remains active for too long, reducing excessive waiting times for competing flows.

The decision rule at each evaluation step is:

- If $k < k_{\min}$, the controller must keep the current phase.
- If $k \geq k_{\max}$, the controller is forced to switch to the opposite phase.
- Otherwise, the controller compares the scores:

- If the current phase is NS green and $S_{EW} > S_{NS}$, it switches to EW green.
- If the current phase is EW green and $S_{NS} > S_{EW}$, it switches to NS green.
- If neither condition holds, the controller keeps the current phase.

This rule-based controller reacts solely to the current observed state and does not maintain any internal belief state or perform learning. It serves as a non-adaptive baseline against which the DQN and AIF controllers can be evaluated.

3.4.5 Deep Q-Network Implementation

The DQN serves as a baseline reinforcement learning approach for traffic signal control. The agent learns to select optimal traffic light phases by approximating the state-action value function $Q(s, a)$ using a neural network.

State and Action Spaces

The state space is a 6-dimensional vector:

$$s_t = [N_{\text{obs}}, E_{\text{obs}}, C_{\text{NS}}, C_{\text{EW}}, B_{\text{NS}}, B_{\text{EW}}]^T \quad (3.12)$$

where $N_{\text{obs}}, E_{\text{obs}}$ are observed vehicle counts, $C_{\text{NS}}, C_{\text{EW}}$ are cumulative CO₂ emissions, and $B_{\text{NS}}, B_{\text{EW}}$ are bus counts.

The action space is discrete with two actions: $a = 0$ for North-South green and $a = 1$ for East-West green.

Reward Function

The reward function balances waiting time minimization, emission reduction, and bus prioritization:

$$r_t = -(W_t + w_c \cdot (C_{\text{NS}} + C_{\text{EW}}) + w_b \cdot (B_{\text{NS}} + B_{\text{EW}})) \quad (3.13)$$

where W_t denotes the total vehicle waiting time. The coefficient w_c controls the contribution of the combined emissions term, scaling it to a comparable magnitude while also reflecting its relative importance in the optimization objective. Similarly, w_b determines the importance assigned to bus prioritization. In this work, the values $w_c = 0.0001$ and $w_b = 0.5$ were selected empirically to achieve stable and balanced controller behavior.

Network Architecture and Training

The Q-network is a feedforward neural network with two hidden layers of 128 neurons each and ReLU activations, mapping the 6-dimensional state to 2 Q-values.

Standard DQN components include:

- **Experience replay:** Buffer of 50,000 transitions, sampling batches of 64
- **Target network:** Updated every 360 steps to stabilize training
- **Discount factor:** $\gamma = 0.99$
- **Learning rate:** 1×10^{-3}

Exploration follows an ϵ -greedy policy with ϵ decaying from 1.0 to 0.01 (decay factor 0.95). Training runs for 100 episodes of 4000 seconds each.

Training Conditions

To reflect realistic training constraints, stochastic disturbances were included during DQN training with limited frequency. Accidents occurred with a probability of 10% and adverse weather with a probability of 40% per episode. This mirrors real-world conditions where such events are irregular and cannot be reliably reproduced during system development. As a result, the DQN agent was not exposed to these disturbances in every training episode.

The loss function minimizes the temporal difference error:

$$\mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} \left[\left(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta) \right)^2 \right] \quad (3.14)$$

3.5 Experimental Setup and Evaluation

Each simulation was run for 3,600 seconds of simulated time. Traffic demand followed a non-stationary pattern, where vehicle arrival rates vary over time, including periods of increased demand and sudden fluctuations. This design enables evaluation of how effectively each controller adapts to changing and unpredictable traffic conditions. To assess robustness, experiments were conducted across multiple scenarios with varying levels of environmental complexity, as described in Section 3.5.1. These scenarios introduce controlled disturbances such as observation noise, adverse weather, and traffic incidents, allowing systematic analysis of controller performance under both realistic and highly challenging conditions. Metrics were collected at one-second intervals using TraCI, ensuring consistent and high-resolution measurement of system behaviour throughout each simulation. The source code and simulation files used for the experiments are provided in Appendix A.

3.5.1 Evaluation Scenarios

To evaluate the robustness and adaptability of the traffic light controllers, experiments are conducted under four distinct simulation scenarios of increasing environmental complexity. These scenarios are designed to systematically introduce uncertainty and disruption into the traffic environment, allowing for controlled comparison under both ideal and realistic operating conditions.

Three sources of disturbance are considered. First, observation noise arises from occlusion effects, where larger vehicles partially obscure smaller ones, leading to imperfect detection of traffic conditions. Second, adverse weather conditions further degrade sensing reliability by amplifying this observation noise. Third, traffic accidents introduce sudden and localized disruptions in traffic flow, creating non-stationary conditions that require rapid adaptation.

These disturbances are implemented using configurable simulation parameters that can be enabled or disabled independently:

- `OCCLUSION_ON`: enables observation noise caused by occlusion effects.

- **WEATHER_ON:** introduces adverse weather conditions, increasing the level of observation noise.
- **ACCIDENT_ON:** introduces stochastic traffic incidents that disrupt normal traffic flow.

Using these parameters, four evaluation scenarios are defined:

- **Scenario 1 (Baseline):** All disturbances are disabled. The environment is fully observable, and traffic conditions evolve deterministically.
- **Scenario 2 (Sensor Noise):** Occlusion is enabled, introducing observation noise. Weather and accidents are disabled. This scenario represents realistic traffic conditions with imperfect sensing.
- **Scenario 3 (Noise + Weather):** Both occlusion and weather effects are enabled, increasing observation uncertainty. Accidents remain disabled.
- **Scenario 4 (Full Disturbance):** All disturbances are enabled, combining observation noise, adverse weather, and traffic accidents. This represents a highly dynamic and partially observable environment.

This structured setup allows for systematic evaluation of controller performance as environmental uncertainty increases, enabling analysis of both efficiency and robustness under progressively more challenging conditions.

4

Results

This chapter presents the empirical results of the simulation study, in which three traffic light controllers, a rule-based baseline, a DQN agent, and an AIF agent are evaluated across four traffic scenarios of increasing environmental complexity. These scenarios range from ideal conditions with perfect information to highly uncertain environments with sensor noise, adverse weather, and traffic disturbances.

The results are reported using four performance metrics: vehicle idle time, CO₂ emissions, bus priority service rate, and phase switching frequency. Particular emphasis is placed on two representative settings: Scenario 2, which reflects realistic traffic conditions with imperfect sensing, and Scenario 4, which represents a worst-case environment with multiple simultaneous disturbances. Scenarios 1 and 3 are included to provide context and illustrate how performance evolves as environmental complexity increases.

4.1 Experimental Conditions

The performance of the evaluated controllers is assessed across four simulation scenarios of increasing environmental complexity. These scenarios introduce controlled variations in sensing uncertainty and traffic disturbances, as described in Section 3.5.1.

The scenarios range from ideal conditions with perfect information to highly dynamic environments with observation noise, adverse weather, and traffic incidents. This setup enables a systematic evaluation of controller performance under both realistic and worst-case conditions.

4.2 Aggregate Performance Across Scenarios

Table 4.1 summarises the performance of all three controllers across the four evaluation scenarios. The results illustrate how controller behaviour changes as environmental complexity increases, from ideal conditions with perfect information to highly uncertain and dynamic traffic environments.

Table 4.1: Aggregate performance across all scenarios. Best values within each scenario are shown in bold.

Scenario	Controller	Avg. Idle (s)	Total CO ₂ (mg)	Bus Rate (%)	Switches
1. Baseline	Rule-Based	38.16	27,589,670	86.69	135
	DQN	31.64	26,872,495	90.91	196
	AIF	26.83	26,746,858	79.11	236
2. Sensor Noise	Rule-Based	40.28	27,799,812	83.62	133
	DQN	30.48	27,138,547	90.24	201
	AIF	22.41	26,669,621	77.13	246
3. Noise + Weather	Rule-Based	41.76	27,973,469	84.19	133
	DQN	29.96	27,267,717	89.23	204
	AIF	26.29	26,563,304	77.05	244
4. Full Disturbance	Rule-Based	125.07	31,051,184	83.67	127
	DQN	244.31	32,691,969	90.57	187
	AIF	67.38	27,333,209	81.57	252

Across all scenarios, the AIF controller consistently achieves the lowest vehicle idle time and CO₂ emissions, indicating superior traffic efficiency. The rule-based controller maintains stable but less efficient behaviour, while the DQN agent exhibits strong performance under simpler conditions but reduced robustness as environmental complexity increases.

Under realistic conditions with sensor noise (Scenario 2), clear performance differences emerge. The AIF agent achieves the lowest idle time, followed by the DQN agent, while the rule-based controller produces higher delays. The differences are moderate, reflecting the relatively controlled nature of this environment.

As uncertainty increases (Scenario 3), performance degrades slightly for all controllers, but the relative ranking remains unchanged. This indicates that the observed differences are consistent across varying levels of sensing uncertainty.

The most pronounced separation occurs in the full disturbance setting (Scenario 4). The DQN agent exhibits a substantial increase in idle time, nearly doubling the delay relative to the rule-based baseline. In contrast, the AIF agent maintains significantly lower idle time, achieving a reduction of approximately 46%. This highlights its ability to adapt to non-stationary and highly stochastic traffic conditions.

Bus priority service rate follows a consistent pattern across all scenarios. The DQN agent achieves the highest prioritisation of bus traffic, while the AIF controller consistently provides the lowest bus service rate. This indicates a trade-off between overall traffic efficiency and explicit prioritisation of specific vehicle classes.

Overall, while all controllers perform well under ideal conditions, their behaviour diverges as environmental complexity increases. AIF provides the most consistent efficiency, DQN offers strong prioritisation but reduced robustness, and the rule-based approach remains stable but less adaptive.

4.3 Vehicle Idle Time

Vehicle idle time, defined as the cumulative time that vehicles remain stationary at the intersection, serves as the primary measure of traffic efficiency. Lower idle time corresponds to reduced congestion and more effective traffic flow.

Across all scenarios, the AIF controller consistently achieves the lowest idle time, followed by the DQN agent, while the rule-based controller produces higher delays. These differences become more pronounced as environmental complexity increases.

Under realistic conditions with sensor noise (Scenario 2), the overall level of congestion remains moderate. Figure 4.1 shows that all controllers exhibit relatively stable behaviour, with limited fluctuations in idle time. However, the AIF agent maintains consistently lower values, while the DQN

agent produces slightly higher variability. The rule-based controller exhibits the highest idle levels throughout the simulation.

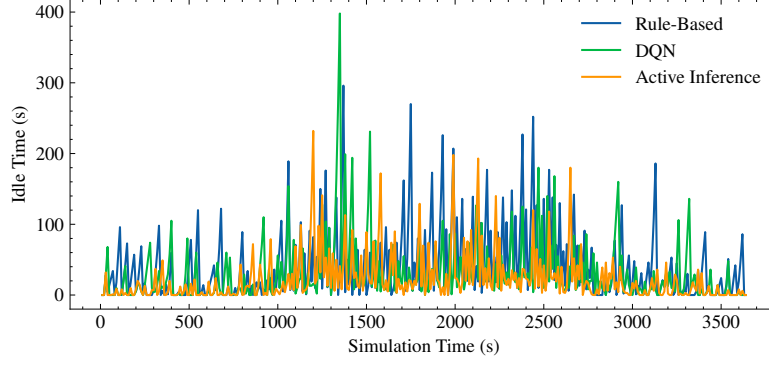


Figure 4.1: Per-step vehicle idle time in Scenario 2 (sensor noise).

This trend is reflected in the cumulative idle time curves in Figure 4.2, where the AIF controller accumulates delay at the slowest rate, followed by the DQN agent and the rule-based baseline.

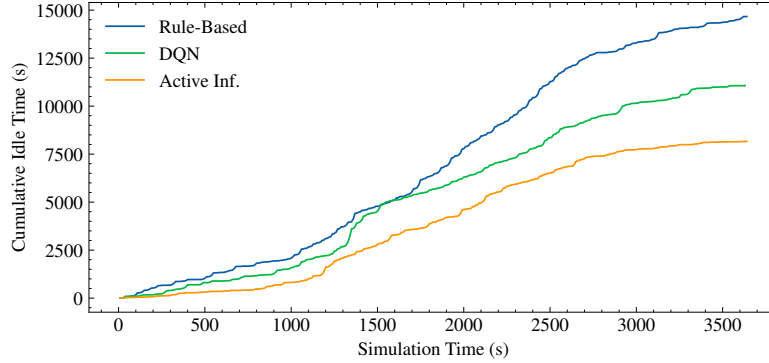


Figure 4.2: Cumulative vehicle idle time in Scenario 2.

The most significant differences arise under the full disturbance setting (Scenario 4). Figure 4.3 shows a sharp increase in idle time during the high-demand period around $t \approx 2000\text{--}2500$ s, corresponding to the introduction of traffic disturbances such as accidents.

The DQN agent exhibits the most extreme spike, indicating severe congestion during this disturbance interval. However, this result should not be interpreted as evidence that DQN generally fails under all noisy or uncertain conditions. Since much of the increase is concentrated around the high-demand disturbance period, the result may partly reflect an out-of-distribution event or insufficient exposure to similar disturbances during

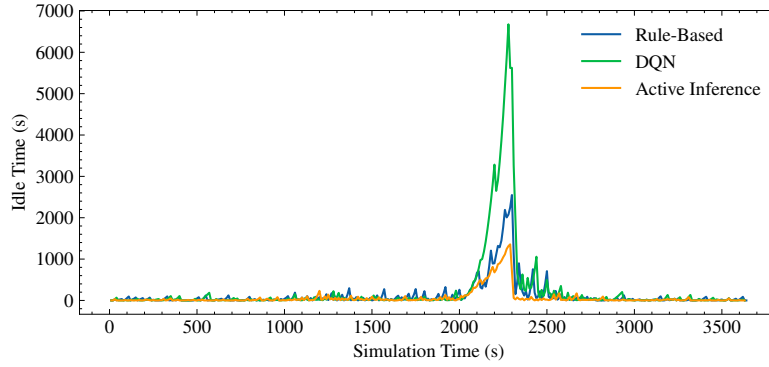


Figure 4.3: Per-step vehicle idle time in Scenario 4 (full disturbance).

training. The rule-based controller shows a more moderate but sustained increase in idle time. In contrast, the AIF agent produces the lowest peak and recovers more rapidly after the disturbance, demonstrating stronger adaptability.

This behaviour is further illustrated in Figure 4.4, where cumulative idle time grows fastest for the DQN agent, followed by the rule-based controller. The AIF agent maintains the slowest accumulation throughout the simulation.

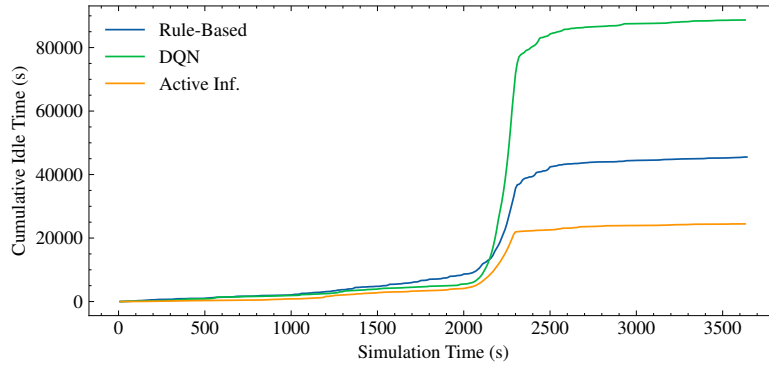


Figure 4.4: Cumulative vehicle idle time in Scenario 4.

An aggregate comparison is provided in Figure 4.5, which shows that the AIF agent achieves the lowest average idle time, while the DQN agent produces the highest delay under full disturbance conditions.

Scenarios 1 and 3 follow similar patterns but with less pronounced differences. Under ideal conditions (Scenario 1), congestion remains low for all controllers, as illustrated by the relatively smooth idle time curves. As uncertainty increases (Scenario 3), idle time rises modestly, but the relative

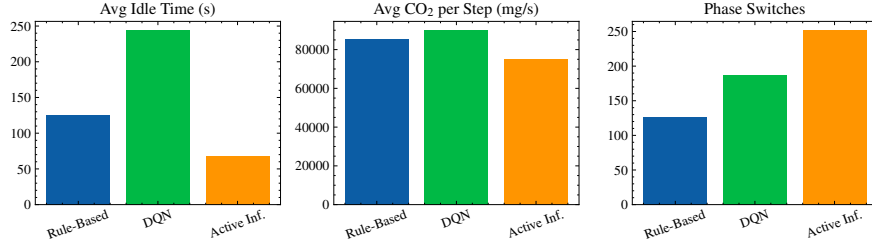


Figure 4.5: Comparison of idle time and related metrics in Scenario 4.

ranking of controllers remains unchanged, with AIF consistently achieving the best performance.

4.4 CO₂ Emissions

CO₂ emissions provide an environmental perspective on traffic efficiency, as higher emissions are typically associated with increased idling and stop-and-go traffic. As a result, the patterns observed in emissions closely follow those of vehicle idle time.

Across all scenarios, the AIF controller consistently produces the lowest CO₂ emissions, while the DQN agent generates the highest values under more complex conditions. The rule-based controller remains intermediate across all settings.

Under realistic conditions with sensor noise (Scenario 2), emissions remain relatively stable over time, as shown in Figure 4.6. All controllers follow a similar overall pattern, reflecting the underlying traffic demand profile. However, the AIF agent consistently maintains slightly lower emission levels, while the DQN agent exhibits higher variability.

In the full disturbance setting (Scenario 4), the differences between controllers become more pronounced. As shown in Figure 4.7, emissions increase sharply during periods of high congestion, particularly around $t \approx 2000\text{--}2500$ s. The DQN agent exhibits the highest peaks, indicating inefficient traffic flow and prolonged idling under disruption.

The rule-based controller shows more moderate but sustained emission levels, reflecting its stable yet less adaptive behaviour. In contrast, the AIF agent maintains lower emission levels throughout the simulation and recovers more quickly after congestion events.

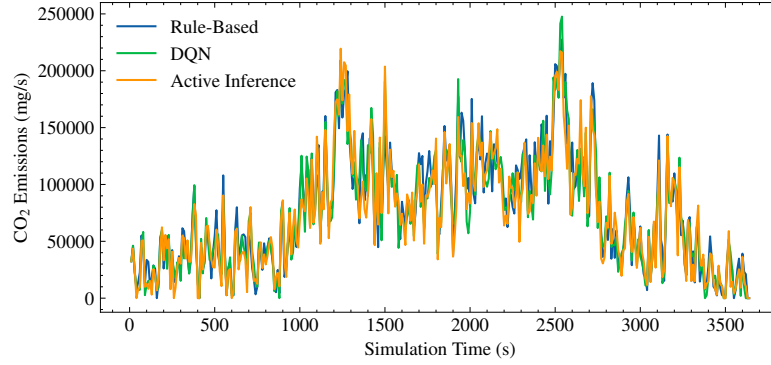


Figure 4.6: Per-step CO₂ emissions in Scenario 2 (sensor noise).

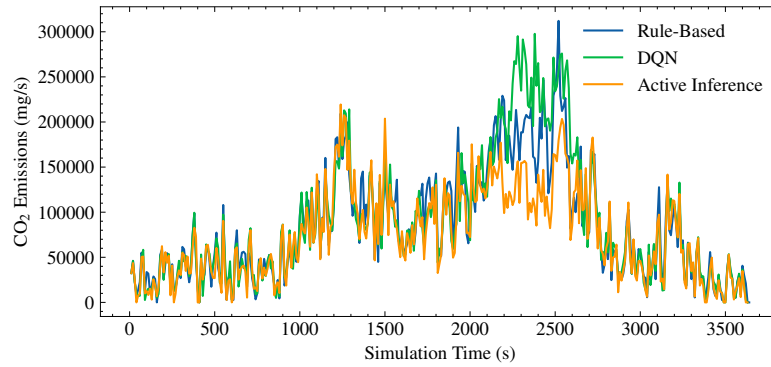


Figure 4.7: Per-step CO₂ emissions in Scenario 4 (full disturbance).

These observations are consistent with the aggregate results reported in Section 4.2, where the AIF controller achieves a reduction of approximately 12% in total CO₂ emissions relative to the rule-based baseline in Scenario 4.

Scenarios 1 and 3 exhibit similar trends but with lower overall emission levels and reduced variability. Under ideal conditions (Scenario 1), emissions remain comparatively low and stable, while under increased uncertainty (Scenario 3), moderate fluctuations emerge without altering the relative ranking of controllers.

4.5 Bus Priority Service Rate

Bus priority service rate measures the proportion of decision steps in which a waiting bus is granted a green signal. This metric reflects how effectively each controller prioritises public transport relative to general traffic flow.

Across all scenarios, a consistent pattern emerges. The DQN agent achieves the highest bus service rate, followed by the rule-based controller, while the AIF agent consistently provides the lowest level of bus prioritisation. This ordering remains stable regardless of environmental complexity.

Under realistic conditions with sensor noise (Scenario 2), the differences between controllers are clearly visible, as shown in Figure 4.8. The DQN agent maintains a high service rate, indicating strong prioritisation of bus traffic. The rule-based controller provides moderate prioritisation, while the active inference agent allocates fewer green phases to buses.

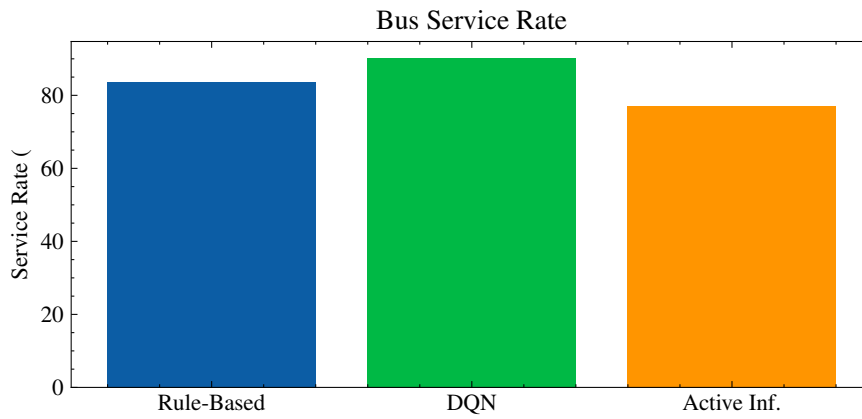


Figure 4.8: Bus priority service rate in Scenario 2.

A similar pattern is observed under full disturbance conditions (Scenario 4), as shown in Figure 4.9. Despite the presence of noise, weather, and traffic incidents, the DQN agent continues to prioritise buses more aggressively than the other controllers.

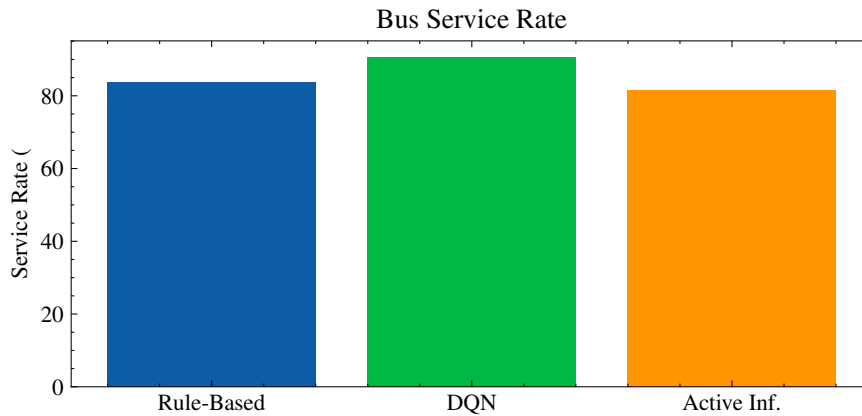


Figure 4.9: Bus priority service rate in Scenario 4.

In contrast, the AIF controller consistently achieves lower bus service rates. This behaviour reflects a different control strategy, where the primary objective is to minimise overall system delay rather than prioritise specific vehicle classes. As a result, bus prioritisation is reduced in favour of improved global traffic efficiency.

The rule-based controller remains intermediate across all scenarios, providing a balance between prioritisation and stability.

Scenarios 1 and 3 exhibit the same relative ordering, with only minor variations in absolute values. Under ideal conditions (Scenario 1), all controllers achieve relatively high service rates, while under increased uncertainty (Scenario 3), small reductions are observed without altering the overall ranking.

Overall, the results highlight a clear trade-off between bus prioritisation and traffic efficiency. The DQN agent maximises service rate at the cost of higher congestion, while the AIF agent achieves lower delays but reduced prioritisation of buses.

The bus priority service rate values in Figures 4.8 and 4.9 are based on the evaluated simulation runs and are therefore reported without confidence intervals. They should be interpreted as scenario-specific outcomes rather than estimates of statistical uncertainty across repeated replications.

4.6 Phase Switch Frequency

Phase switch frequency measures how often a controller changes the traffic signal phase. This metric provides insight into the stability and responsiveness of each control strategy. Frequent switching may indicate high reactivity, while fewer switches suggest more stable but potentially less adaptive behaviour.

Across all scenarios, clear differences in switching behaviour are observed between controllers. The rule-based controller consistently produces the lowest number of phase switches, indicating a stable and predictable control policy. The DQN agent exhibits a higher switching frequency, reflecting a more dynamic but less stable strategy. The AIF controller consistently produces the highest number of switches, indicating a highly reactive approach.

Under realistic conditions with sensor noise (Scenario 2), these differences are clearly visible in Figure 4.10. The rule-based controller maintains a relatively steady switching pattern over time. The DQN agent shows greater variability, with periods of increased switching activity. The AIF controller frequently changes phase, often at nearly every decision step.

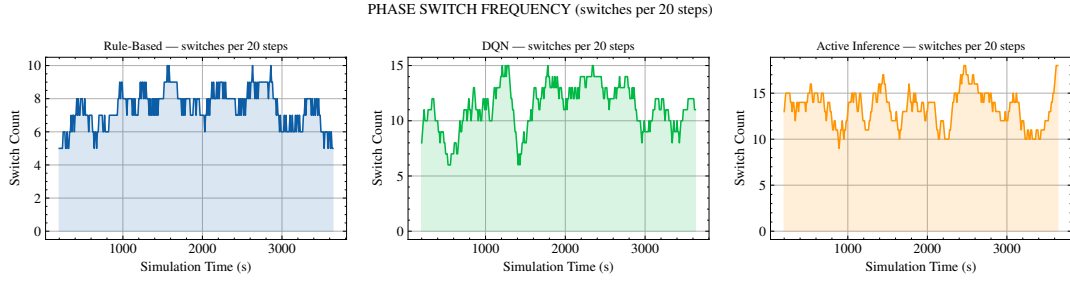


Figure 4.10: Phase switching frequency in Scenario 2 (sensor noise).

A similar pattern is observed under full disturbance conditions (Scenario 4), as shown in Figure 4.11. Despite the increased environmental complexity, the relative ordering of controllers remains unchanged. However, the variability of switching behaviour increases, particularly for the DQN agent, which exhibits irregular switching patterns under disturbance.

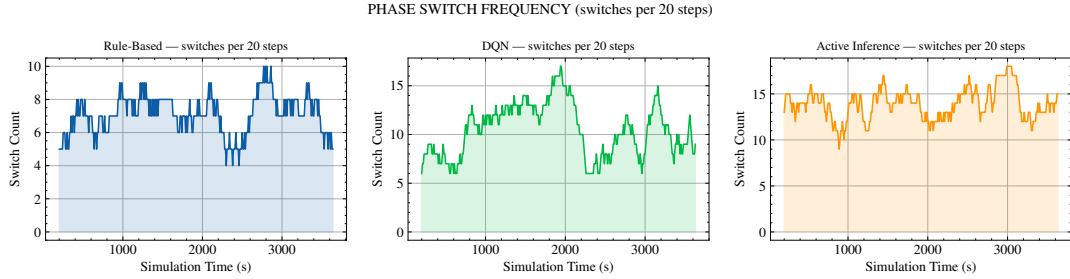


Figure 4.11: Phase switching frequency in Scenario 4 (full disturbance).

The high switching frequency of the AIF controller helps explain its strong performance in terms of idle time and emissions. By reacting quickly to changing traffic conditions, it is able to redistribute green time more effectively. However, this behaviour comes at the cost of reduced stability and may contribute to lower bus priority service rates, as observed in Section 4.5.

In contrast, the rule-based controller's low switching frequency results in stable but less responsive behaviour, leading to higher idle times under dynamic conditions. The DQN agent occupies an intermediate position,

balancing responsiveness and stability but struggling to maintain consistent performance under high levels of uncertainty.

Scenarios 1 and 3 exhibit the same overall ordering, with lower variability under ideal conditions (Scenario 1) and slightly increased fluctuations under moderate uncertainty (Scenario 3). These results indicate that switching behaviour is an inherent characteristic of each controller, largely independent of the specific scenario.

4.7 Scenario-Specific Dynamics

To better understand how the evaluated controllers respond to non-stationary traffic conditions, this section examines the temporal dynamics of Scenario 4, which represents the most challenging environment with sensor noise, adverse weather, and traffic disturbances.

Figure 4.12 illustrates the evolution of vehicle idle time over the simulation horizon. All controllers experience a significant increase in congestion during the high-demand period around $t \approx 2000\text{--}2500$ s. This period corresponds to the introduction of disturbances such as traffic accidents, which disrupt normal traffic flow and create localized bottlenecks.

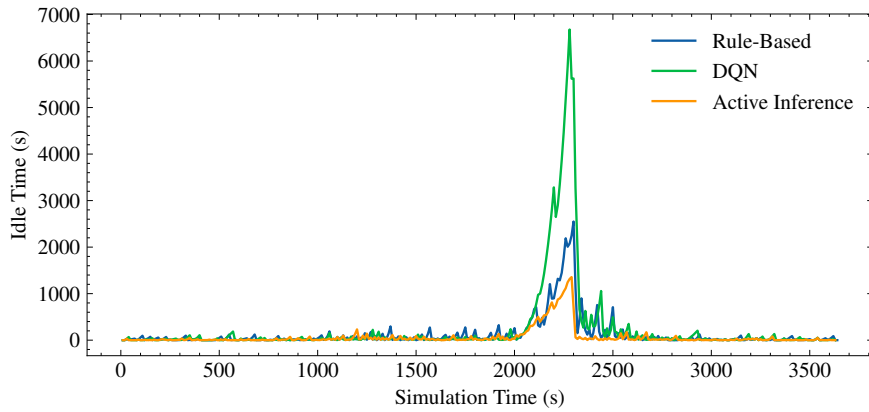


Figure 4.12: Vehicle idle time over time in Scenario 4.

The magnitude and duration of congestion differ substantially across controllers. The DQN agent exhibits the largest spike in idle time, indicating severe congestion and an inability to adapt effectively to sudden disruptions. The rule-based controller shows a more moderate but sustained increase in delay, reflecting its stable but less responsive behaviour.

In contrast, the AIF controller produces lower peak values and demonstrates faster recovery following the disturbance. This suggests that its reactive decision-making process enables more effective redistribution of green time under rapidly changing traffic conditions.

The cumulative effect of these dynamics is shown in Figure 4.13. The DQN agent accumulates idle time at the fastest rate, followed by the rule-based controller, while the AIF agent maintains the lowest cumulative delay throughout the simulation.

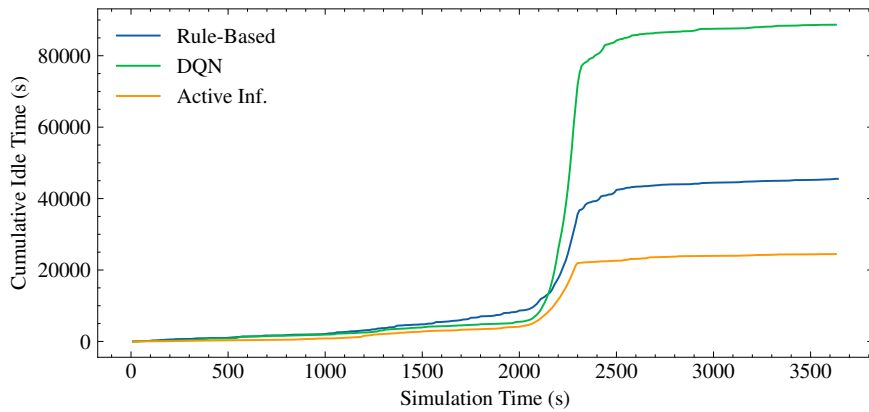


Figure 4.13: Cumulative idle time in Scenario 4.

A similar pattern is observed in CO₂ emissions, where periods of increased congestion correspond to elevated emission levels. As shown in Figure 4.14, the DQN agent produces the highest emission peaks during disturbance periods, while the AIF controller maintains lower levels and recovers more quickly.

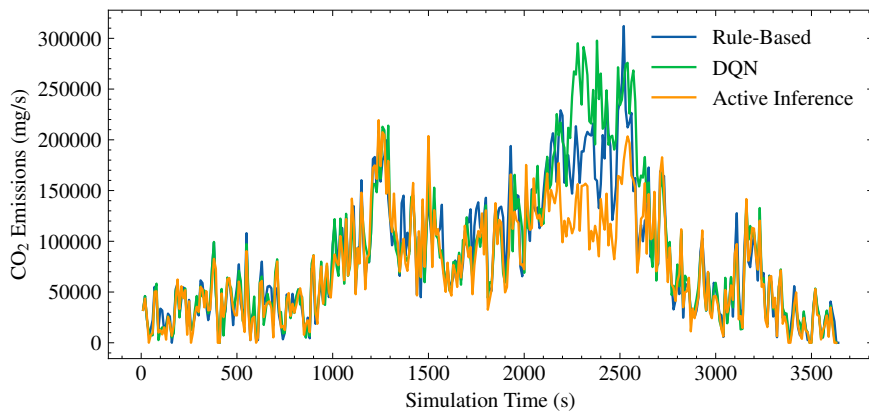


Figure 4.14: CO₂ emissions over time in Scenario 4.

These results highlight the importance of adaptability in non-stationary environments. Controllers that respond quickly to changes in traffic conditions are better able to mitigate the impact of disruptions, reducing both congestion and emissions. In contrast, less adaptive strategies either over-react or fail to respond appropriately, leading to sustained inefficiencies.

Overall, the temporal analysis of Scenario 4 reinforces the findings from previous sections: the AIF controller provides the most robust performance under dynamic and uncertain conditions, while the DQN agent struggles to maintain stability, and the rule-based controller remains stable but less efficient.

4.7.1 Interpretability of Active Inference

A key qualitative aspect of the AIF framework is its interpretability, as the decision-making process is explicitly grounded in a structured generative model. Unlike black-box approaches, the model separates perception, dynamics, and action selection into distinct and inspectable components, allowing the reasoning process to be traced step by step.

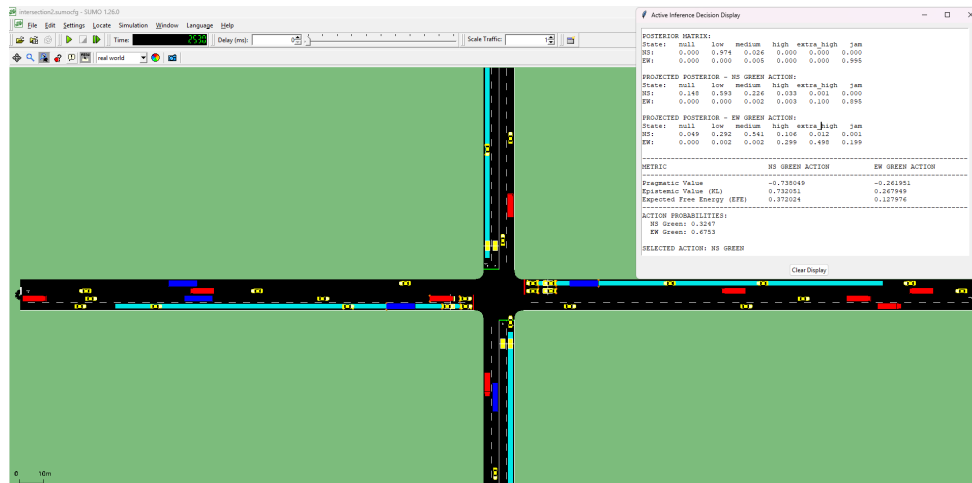


Figure 4.15: Illustration of a single decision step in the AIF framework applied to traffic light control.

The figure illustrates how a single control decision is generated within a unified probabilistic framework, where each stage corresponds to an explicit quantity under the model's internal beliefs.

First, observations from the traffic environment are used to update posterior beliefs over hidden states, yielding an inference about the current latent situation conditioned on sensory evidence. These inferred state distributions are then propagated through the transition model to form predictive beliefs about the outcomes of alternative traffic signal phases.

Each candidate action is evaluated by computing its expected free energy, which integrates predicted outcome preferences with expected uncertainty reduction under the model's generative assumptions. This yields a principled trade-off between exploitation and information gain.

Finally, actions are assigned probabilities proportional to their expected free energy, producing a posterior distribution over control decisions.

Importantly, perception, prediction, and action selection are all expressed in terms of explicit probabilistic quantities within the same generative model, providing a fully structured and transparent inference-to-decision pipeline.

5

Discussion

5.1 Answer to the Research Questions

RQ1: Performance compared to existing methods

The results show that the AIF controller performed best on the main efficiency metrics. Across all four scenarios, it achieved the lowest average vehicle idle time and the lowest total CO₂ emissions. This indicates that, in the tested simulation environment, AIF improved traffic flow and reduced emissions compared with the rule-based controller and the DQN agent.

However, AIF did not perform best on all metrics. The DQN agent achieved the highest bus priority service rate in all scenarios, while the rule-based controller had the lowest phase switching frequency. Therefore, the answer to RQ1 is that AIF outperformed the other evaluated controllers in terms of idle time and CO₂ emissions, but not in terms of bus priority or signal stability. The comparison therefore shows a trade-off: AIF improved overall efficiency, while DQN prioritised buses more strongly and the rule-based controller produced more stable signal switching.

RQ2: Adaptation under non-stationary and uncertain conditions

The results indicate that AIF handled increasing uncertainty better than the DQN agent and the rule-based controller in the tested scenarios. This is most visible in Scenario 4, where observation noise, adverse weather, and accidents were combined. In this scenario, the AIF controller maintained lower idle time and lower CO₂ emissions than both comparison methods.

The DQN agent performed well under less complex conditions, but its idle time increased substantially under full disturbance conditions. This should be interpreted cautiously, because the degradation in Scenario 4 appears to be concentrated around a severe disturbance interval rather than distributed evenly across the full simulation. Therefore, the result may indicate limited robustness to an out-of-distribution traffic event in this experiment, rather than a general inability of DQN controllers to handle uncertainty. The rule-based controller remained more stable in terms of phase switching, but it produced higher idle time and emissions than AIF. Therefore, the answer to RQ2 is that AIF showed better adaptation to the simulated uncertain and non-stationary conditions, especially when disturbances were combined. This conclusion is based on the observed idle time and CO₂ results, not on a separate real-world robustness test.

RQ3: Trade-offs of using Active Inference

RQ3 investigated what trade-offs arise when using AIF in terms of interpretability, computational feasibility, and suitability for real-world deployment. Regarding interpretability, the AIF controller uses explicit model components, including hidden states, transition probabilities, preferences, and belief updates. This means that at every decision step it is straightforward to inspect why the agent selected a particular action and what factors contributed to that decision, which is in contrast to the DQN baseline where policies are represented as implicit mappings from observations to actions.

In terms of computational feasibility and real-world suitability, AIF requires no gradient-based optimisation, large training data, or meticulous reward engineering, making it lighter and more straightforward to configure than the DQN agent. It achieved competitive performance without extensive tuning, suggesting a practical advantage in settings where data is

limited or extensive optimization is not possible. However, further validation beyond a single simulated intersection is needed before stronger conclusions about real-world deployment can be drawn.

One practical concern, however, is the AIF controller’s high phase switching frequency, approximately twice that of the rule-based controller. While this reactivity reduces idle time, it conflicts with minimum green time requirements in real traffic systems and raises concerns about mechanical wear and driver predictability.

5.2 Comparison with Previous Research

AIF achieves performance comparable to established control methods when parameters are tuned appropriately, consistent with prior results in other domains. Lanillos et al. [24] review AIF applications in robotics and artificial agents, highlighting its use for state estimation, control, planning, learning, adaptation, generalization, and robustness. In traffic signal control, Gao et al. [16] reported that a DQN-based controller reduced vehicle delay by up to 47% relative to a longest-queue-first policy and 86% relative to fixed-time control. The present results place AIF within a similar performance range without requiring an explicitly specified reward function or gradient-based optimisation.

AIF maintains performance under partial observability because it models the control problem as a partially observable Markov decision process. Malekzadeh and Plataniotis [29] showed that minimising EFE drives both reward-seeking and information-seeking action selection. Standard reinforcement learning methods optimise reward estimates and typically treat state uncertainty as noise rather than as a quantity to reduce. This distinction matters in real deployments: Chen et al. [4] identified limited sensing and non-stationary traffic patterns as primary causes of performance degradation when transferring RL-based controllers from simulation to physical systems.

AIF exposes internal variables, such as beliefs over states and prior preferences, at each decision step, whereas DQN represents policies as implicit mappings from observations to actions. Liao et al. [27] argue that neural traffic-signal-control policies are often over-parameterized and non-transparent, which can limit interpretability and deployability. By contrast,

AIF provides direct access to the quantities that determine action selection, which enables inspection without post-hoc analysis.

Performance depends on correct parameter settings. In AIF, a precision parameter controls how strongly the agent favors higher-valued policies during action selection. Friston et al. [15] discuss how precision shapes confidence in policy and action selection. Similar effects appear in traffic signal control more broadly: Wang et al. [46] examine how varying intervention frequencies affect traffic signal cycle control, indicating that timing and intervention settings can influence controller performance.

5.3 Limitations

The results of this study should be interpreted within the boundaries of the tested simulation setting. While the comparison between controllers is internally consistent, several factors limit how far the conclusions can be extended.

The simulation environment approximates but does not fully replicate real traffic conditions. Although SUMO enables controlled evaluation, the disturbance mechanisms representing observation noise, adverse weather, and accidents do not capture the temporal continuity, spatial variation, or correlated uncertainty found in real urban traffic systems. Generalization to real infrastructure therefore requires further validation in larger and more realistic networks.

The AIF controller’s performance is also sensitive to its model specification. The use of discretised hidden states, manually defined transition probabilities, predefined preference structures, and fixed precision parameters makes the controller inspectable and computationally manageable, but means that different modelling choices could produce meaningfully different behaviour.

A further consideration is that the controllers were not evaluated under identical phase-switching constraints. The AIF controller switches signal phases more frequently than the other two, meaning some performance differences may partly reflect switching frequency rather than the quality of the control strategy alone.

The stochastic nature of traffic arrivals, observation noise, and random disturbances also introduce variation across runs. Reproducibility depends on access to implementation details, parameter settings, and random seeds. Computational constraints additionally required coarse state discretisation and a short policy horizon, which may have affected AIF performance.

Overall, the results show that AIF performs well in the tested scenarios, particularly in terms of vehicle idle time and CO₂ emissions, but do not establish how it would perform in larger networks, under different demand patterns, or in real-world deployment.

5.4 Validity, Reliability, Generalizability, and Objectivity

The methodological quality of this study is discussed in terms of internal validity, external validity, reliability, and objectivity, which concern whether the study measures what it intends to measure, whether findings can be generalized, whether results can be reproduced, and whether the analysis is systematic rather than subjective [6, 38].

All controllers were evaluated under identical traffic demand profiles, disturbance configurations, simulation duration, and measurement procedures, which supports internal validity. This is limited, however, by stochastic elements such as random traffic arrivals, observation noise, and accident events, as well as the fact that controllers were not constrained to identical phase-switching behaviour.

The study was conducted in SUMO using a simplified four-arm intersection with predefined scenarios and simulated disturbances, which limits direct generalization to real-world systems, larger networks, and different intersection layouts. Further studies are required before conclusions can be extended to operational traffic management.

Reliability is supported by defined simulation scenarios, fixed controller configurations, and consistent performance metrics. Because the simulation contains stochastic components, transparent reporting of implementation details, parameter settings, and random seeds is important for reproducibility.

Quantitative metrics logged automatically from the simulation support objectivity, though the selection of metrics inevitably influences how performance is interpreted. AIF performs best in terms of idle time and CO₂ emissions, while DQN leads on bus priority service rate, meaning conclusions depend on which control objectives are considered most important.

5.5 Future Work

Future work should focus on AIF for traffic light control rather than on AIF or traffic control in isolation. A central direction is more systematic specification and tuning of the AIF controller. The controller depends on several modelling choices, including hidden-state definitions, transition probabilities, preference structures, epistemic weighting, policy precision, and state discretisation. These parameters interact in non-linear ways, which makes manual tuning difficult. Sensitivity analysis, automated parameter search, or data-driven estimation of parts of the generative model could help identify which modelling choices most strongly affect controller behaviour and how the controller can be configured more consistently across traffic scenarios.

Another important direction is evaluation in more realistic traffic environments. The present study uses a controlled SUMO simulation with a single four-arm intersection. This setting supports controlled comparison, but it does not represent larger road networks, different intersection geometries, coordinated signal systems, or real-world infrastructure constraints. Future studies should therefore test the controller in multi-intersection networks, using empirical traffic data, turning movements, pedestrian phases, safety constraints, and more realistic demand patterns. This would make it possible to examine whether the reductions in vehicle idle time and CO₂ emissions observed in this thesis remain consistent under more complex conditions.

The comparative evaluation should also be expanded. This thesis compares active inference with a rule-based controller and a DQN agent. Later studies could include additional adaptive traffic-control methods, such as model predictive control, Proximal Policy Optimization, Advantage Actor-Critic, and multi-agent reinforcement learning approaches. These comparisons should include systematic parameter tuning for all methods where

tuning is applicable, so that performance differences can be interpreted more fairly. This is especially important because the present study evaluates fixed controller configurations rather than fully optimised versions of each method.

Overall, future research should move from proof-of-concept simulation toward a more systematic evaluation of active inference for traffic light control. This requires better parameter-tuning procedures, more realistic traffic scenarios, and broader comparisons with established adaptive control methods.

5.6 Ethical and Societal Implications

The societal relevance of this thesis follows from the problems discussed in the introduction: congestion can contribute to air pollution, reduced mobility, economic loss, and accident risk [26, 35, 42, 45]. The results indicate that the active inference controller reduced vehicle idle time and CO₂ emissions in the evaluated simulation scenarios. If similar effects could be achieved in real traffic systems, adaptive traffic control could contribute to lower congestion and environmental impact.

However, traffic-control objectives involve social trade-offs. In this study, the active inference controller performed best on idle time and CO₂ emissions, while the DQN agent achieved the highest bus priority service rate. This shows that different control objectives may benefit different aspects of the traffic system. A real deployment would therefore need to make priorities explicit, for example whether the system should prioritise average vehicle flow, public transport, emissions, pedestrians, or safety.

Because traffic lights are public infrastructure, automated controllers require careful validation before deployment. This thesis evaluates active inference only in simulation, so the results are not sufficient for practical use. Future work should test the controller in more realistic networks and examine how different road-user groups are affected.

5.7 Use of AI Tools in the Thesis Work

AI-based tools, including Microsoft Copilot, Claude, and ChatGPT, were used in a supportive capacity during the thesis work. Their use focused on improving clarity, refining academic language, and assisting with literature search tasks. All conceptual development, methodological decisions, implementation work, and analytical reasoning were carried out independently by the authors.

AI tools did not generate research results, perform data analysis, or replace the authors' reasoning at any stage. Their role was limited to auxiliary writing, editing, and information-support functions.

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Appendix A

Source Code Repository

The source code used for the implementation and evaluation of the traffic light controllers is available in the following GitHub repository:

`https://github.com/GMAN226/Python-Code-Active-Inference-for-Intelligent-Traffic-Light-Control`

The repository contains the Python code, SUMO related files, controller implementations, and evaluation scripts used during the thesis work. This includes the AIF controller, the DQN baseline, the rule-based controller, and scripts for collecting and analysing simulation outputs.

To support reproducibility, the repository corresponds to the implementation used to produce the results reported in this thesis.

Appendix B

Reflection Document George Ambroladze

The thesis fits well with the goals of the thesis course. It particularly emphasizes the need to conduct independent academic work and contribute to knowledge development in computer and systems sciences. The thesis required us to identify a relevant research problem, review previous studies, design and implement an artefact, evaluate it through simulation, and critically discuss the results. This work was both a theoretical study and a practical implementation of an Active Inference controller for traffic light control. As a result, the thesis was strongly connected to design research and empirical evaluation.

We met the course objectives of selecting and applying scientific methods, using relevant literature, and writing and defending an academic report. We adopted Design Science Research as our overall research strategy, as the thesis involved developing and evaluating an artefact. We also used simulation-based experiments in SUMO to compare the Active Inference controller with a rule-based controller and a DQN agent. This approach helped us answer the research questions in a structured manner. I believe the thesis contributed to knowledge development because Active Inference remains relatively unexplored in traffic light control compared to reinforcement learning approaches. However, the generalizability of the

results was less well met. The thesis focused on a simulated four-arm intersection, which means the results cannot be directly applied to real-world traffic systems or larger road networks without further testing.

In general, the planning of the thesis went smoothly. We held bi-weekly meetings with our supervisor, Alfreds, which helped us stay on track throughout the process. These meetings provided a clear structure and made it easier to identify areas for improvement after each draft. Time management was not a major issue, and the work was divided fairly evenly between my thesis partner and me. However, the coding turned out to be more challenging than I expected. We spent a lot of time writing the first three chapters early on, and later I realized that implementing and tuning the parameters needed more attention than I had anticipated. In hindsight, we could have started coding and experimenting earlier and focused the early theoretical chapters more on the environment, methods, and research setup.

The most challenging part of the thesis was implementing and tuning the controllers. The DQN agent required hyperparameter tuning, and the Active Inference controller needed careful parameter selection to fit the simulation model. These choices influenced the final behaviour of the controllers, so it was not just a technical coding task but also an analytical one. We had to understand how the parameters affected traffic flow, idle time, emissions, bus service rate, and phase switching. Given more time, I would have liked to test the controllers in more scenarios with different traffic setups. However, this would have extended the thesis and made it more complex, so I believe the final scope was reasonable for a master's thesis.

The thesis closely relates to my education in Data Science, Statistics, and Decision Analysis at Stockholm University. The most relevant areas were Python programming and reinforcement learning. Python was crucial for coding, running simulations, collecting results, and creating plots. Understanding reinforcement learning was important because the DQN agent was one of the main comparators. This knowledge also helped me compare it fairly with Active Inference. Statistics and data analysis were relevant since we needed to collect, visualize, compare, and interpret the results. More broadly, the thesis integrated many aspects of the programme: data science, AI, simulation, decision-making under uncertainty, and analytical reasoning.

The thesis is valuable for my future work and studies because it provided me with experience in a complete research and development process. I did not just apply existing code or analyze a pre-made dataset; I helped build a simulation environment, implement controllers, run experiments, analyze results, and present conclusions. This experience is useful for future work in data science, AI, analytics, and decision support, as many real problems require both technical implementation and critical interpretation. The project also increased my confidence in working with complex systems where no single correct answer exists and where results depend on modelling assumptions, parameter selections, and evaluation designs.

I am very satisfied with how the thesis was executed and its outcome. The process was demanding, especially during the implementation and revision stages. However, I believe the final result is much stronger because of that. Alfreds provided detailed feedback, and though the revisions took time, they significantly improved the quality of the thesis. I am also pleased with the clear results: the Active Inference controller performed well regarding idle time and CO₂ emissions, while the DQN agent excelled in bus priority service rate. This allowed us to draw a balanced conclusion rather than simply claiming one method was superior in every respect. I take pride in our supervisor being satisfied with the final work and that we are moving toward publication.

During the thesis process, we used AI tools like ChatGPT, Claude, and Microsoft Copilot as support. These tools mainly helped improve paragraph clarity, check structure, summarize feedback, and assist with debugging code. For instance, AI tools were beneficial when we wanted to refine a paragraph or when we needed help figuring out why a piece of Python code was not functioning as expected. They also helped organize supervisor feedback into specific revision tasks. However, the AI tools did not generate research results, make methodological decisions, or replace our reasoning. We had to review their suggestions carefully, particularly regarding technical and academic accuracy. Overall, the tools effectively supported us, especially when I already understood the problem and could evaluate whether a suggestion made sense.

Appendix C

Reflection Document

Dénes Tóth

The Master's program in Data Science, Statistics, and Decision Analysis at Stockholm University aims to develop independent research capacity, applied analytical skills, and the ability to communicate findings in a scientific context. This thesis addressed all of these areas. The project required us to independently identify a gap in existing research, build a theoretical framework around it, implement a working solution, and evaluate it through controlled experiments. Active inference as a control framework sits directly at the intersection of probabilistic modelling and decision-making under uncertainty, both of which are central themes throughout the program.

The thesis also met the course requirement of contributing to knowledge development. Active inference has been applied in robotics and neuroscience but remains largely unexplored in traffic signal control, so the study adds something new rather than simply replicating existing work. Where the thesis fell slightly short of the program goals was in generalizability. The evaluation was limited to a single simulated intersection, which makes it harder to draw broad conclusions. This is acknowledged in the thesis but is still a limitation worth noting here.

The planning phase worked reasonably well. We had bi-weekly meetings with our supervisor which gave the project a clear rhythm and made it easier to catch problems early. The research plan was a useful reference point throughout, though it did not fully account for how time-consuming the implementation stage would turn out to be. Looking back, I spent too much time on the theoretical chapters early on and should have started experimenting with the simulation environment sooner. Getting familiar with SUMO and tuning the controllers ended up taking significantly more time than I had planned for, which created pressure later in the process.

The workload overall corresponded well to what a master's thesis should require. The conditions were good and supervision was helpful and consistent.

The execution stage was the most demanding part of the project. Implementing the active inference controller was not just a coding task. It required understanding how each parameter affected the behavior of the agent and how changes in the model influenced traffic outcomes like idle time, emissions, and phase switching. This analytical side of the implementation took longer than expected and I underestimated it at the start.

What went better than expected was the quality of the results. The active inference controller performed consistently well across scenarios, particularly under the full disturbance condition, which made the results easier to interpret and discuss. I had expected the comparison to be less clear-cut.

Writing the report itself was manageable. I felt prepared enough from previous academic work to structure a thesis, though the revision process took more time than I anticipated. Supervisor feedback was detailed and genuinely improved the quality of the writing, even when incorporating it required reworking sections I thought were finished. My English was sufficient throughout though some sections needed extra editing passes to reach the right academic tone.

I am satisfied with the final result. Given the complexity of building and evaluating three different controllers in a simulated environment, I think the thesis ended up being thorough and well-argued.

The most useful parts of my education were machine learning, Python programming, and statistics. Reinforcement learning knowledge was essential for implementing and understanding the DQN baseline and for contextualizing active inference within the broader AI landscape. Python was used throughout for everything from running simulations to collecting results and generating figures. Statistical thinking was important when interpreting results and understanding what the numbers meant beyond surface comparisons.

Active inference was not covered during the program so a significant part of the thesis involved independent study of the theoretical neuroscience and computational modelling literature behind it. Understanding the free energy principle and expected free energy well enough to implement them correctly required reading well outside standard data science material. Working with SUMO was also entirely new and took time to learn before meaningful experiments could be run.

This project gave me experience with a full research and development cycle, from problem definition through implementation to evaluation and written communication. That process, where there is no predefined answer and where results depend on modelling choices and evaluation design, reflects what real work in data science and AI actually looks like. I feel more confident going into that kind of work having completed a project of this scope. The fact that the thesis is moving toward publication is also encouraging and suggests the work has value beyond the degree itself.